Chapter 5

离子注入

Ion implantation

李秀妍

xiuyanli@sjtu.edu.cn

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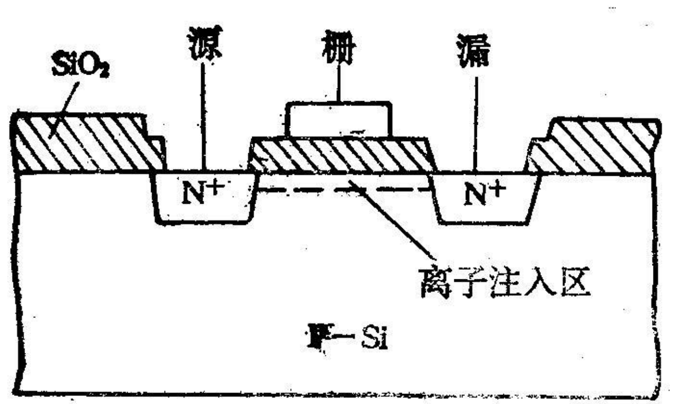
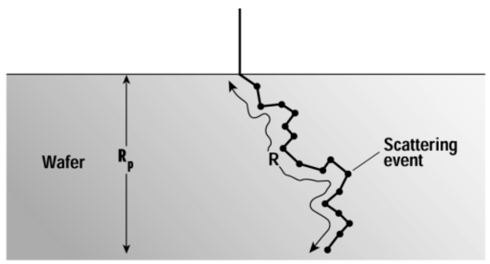
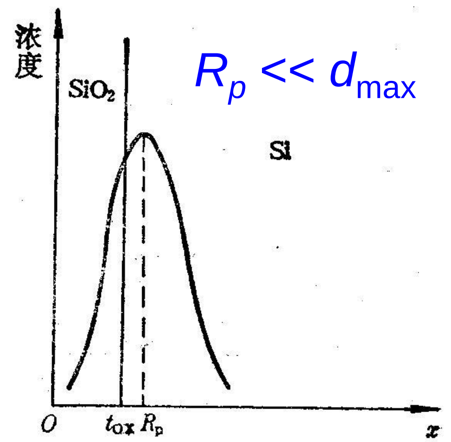
Outline

5.1 Key reason to introduce





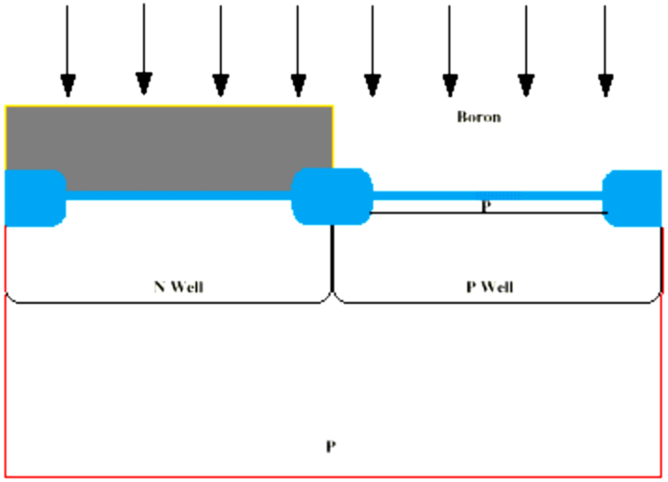
$$V_T = \boxed{\phi_{ms}} - \frac{Q_{ox}}{C_{ox}} + 2\psi_b + \frac{\sqrt{2\varepsilon_s q N_A (2\psi_b)}}{C_{ox}}$$



$$Q_B^{total}(d_{\max}) = \int_0^{d_{\max}} q\left[N_A + N_A^{'}(x)\right]dx = Q_B(d_{\max}) + \Delta Q_B(d_{\max})$$

$$\Delta Q_B(d_{\max}) = \int_0^{d_{\max}} qN_A^{'}(x)dx = qN_{Im} \qquad \Delta V_T = \frac{\Delta Q_B}{C_{ox}} \approx \frac{qN_{Im}}{C_{ox}}$$

5.1 Key reason to introduce



NMOS V_{TH} adjust: B is ion implanted to adjust V_{TH}.

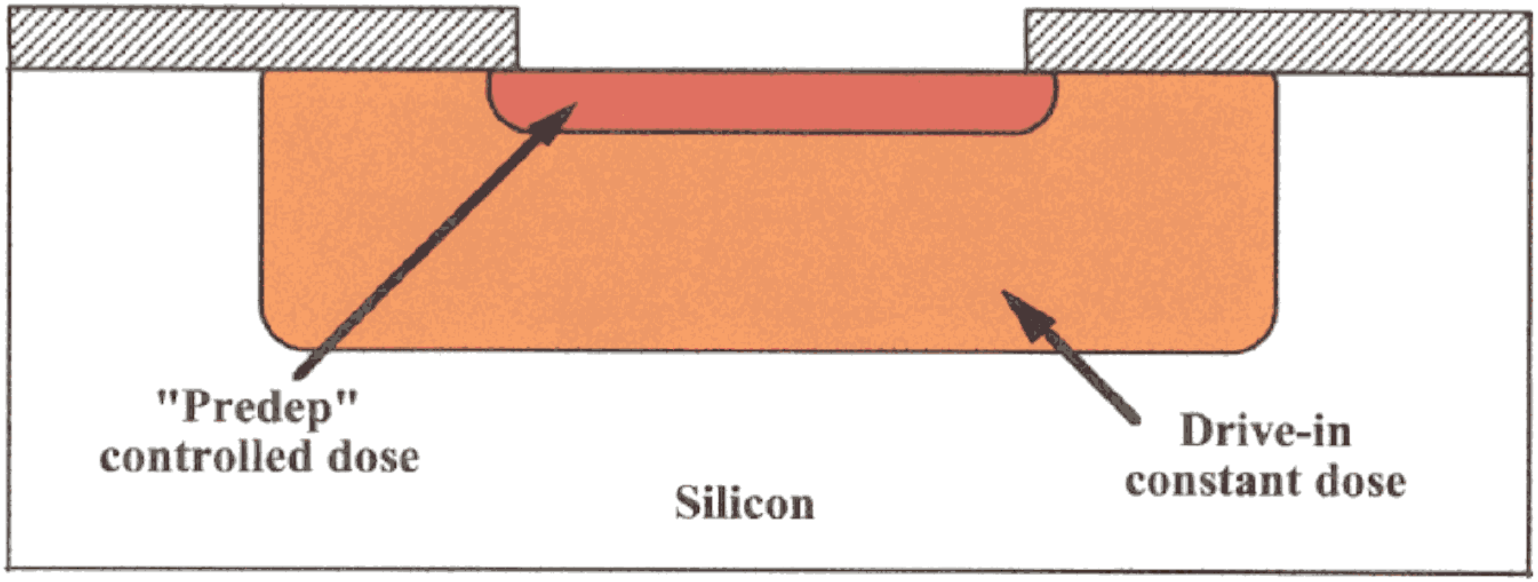
Typical conditions:
Dose = 1- 10 x 10^{12} cm^{-2}
Energy ~ 50 KeV

Dose,ϕ, can be accurately controlled because it is based upon integrating the ion beam current into the wafer

$$\Phi = 1/A \int (I/q)\, dt$$

A = implant area
I = collected beam current (from back of wafer)
t = integration time
q = charge on ion

5.1 Application in modern predeposition



Historical development: original method of introducing dopants into the silicon crystal was by doing a gas-phase diffusion (high temperature) masked by oxide. This method has serious limitations on the control of the dose of dopant introduced, as well as the profile shape.

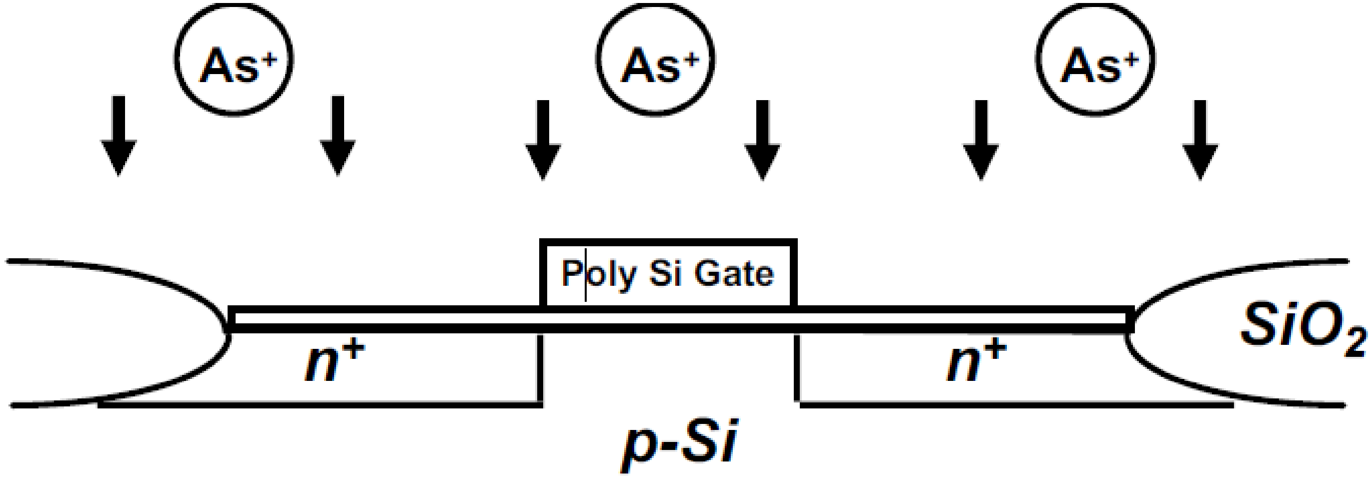
Modern method: use ion implantation to introduce dopants, and annealing/diffusion to repair damage and, if necessary, diffuse the profile.

5.2.1 Advantages of ion implantation

- Precise control of dose and depth profile
- Low--temperature process
- Wide selection of masking materials

  e.g. photoresist, oxide, poly--Si, metal
- Excellent lateral uniformity (< 1% variation across 12” wafer)
- Less sensitive to surface cleaning procedures

Application example: self--aligned MOSFET source/drain regions



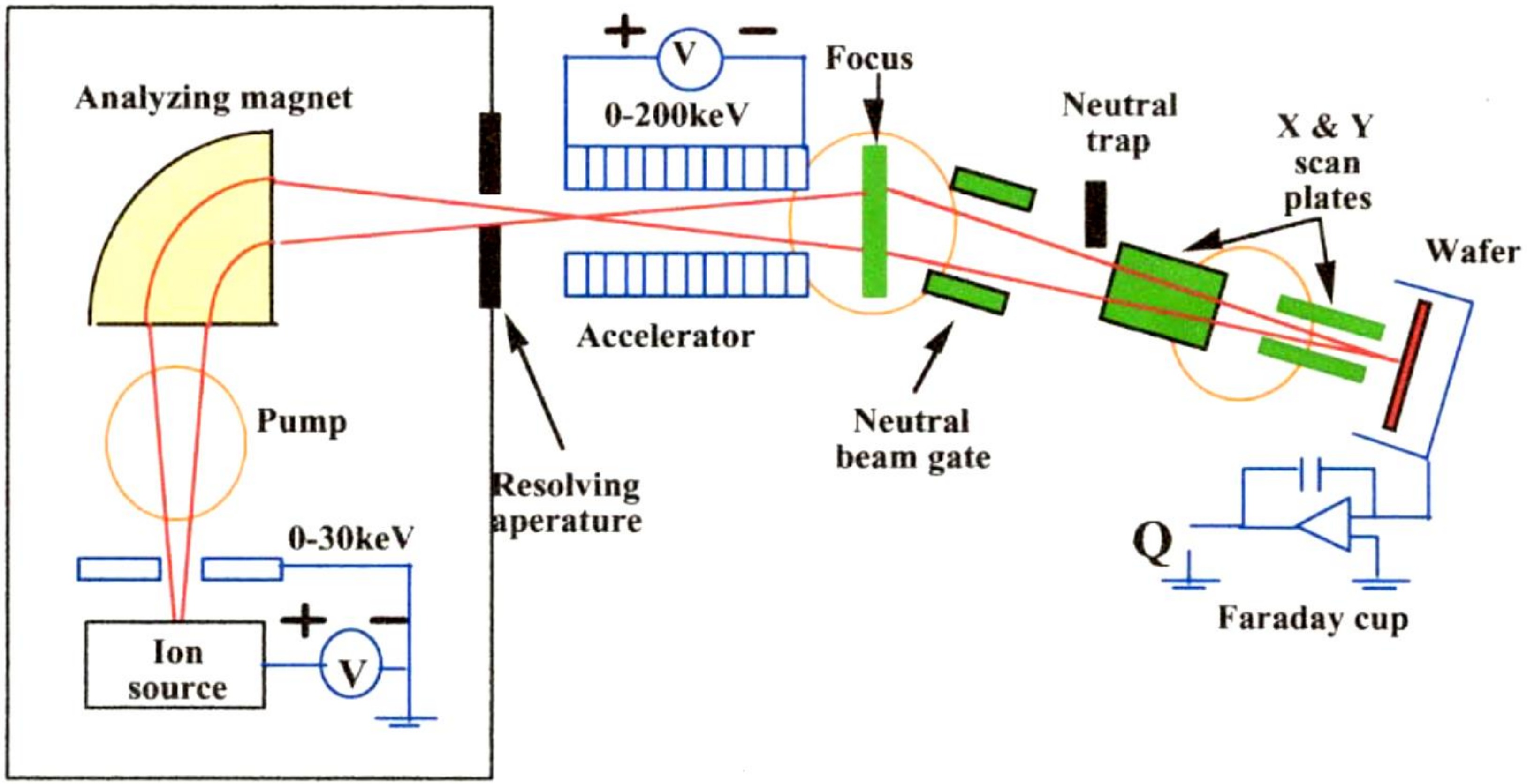
Outline

- 5. 1. Why is ion implantation needed?
- 5. 2. Basics of ion Implanter
- 5. 3. Basics of ion implantation
- 5. 4. Selective implantation
- 5. 5. Other aspects of ion implantation
- 5. 5. Summary

5.2.1 Overview of ion implantation

- Wafer is target in high energy accelerator
- Impurities "shot" into wafer
- Preferred method of adding impurities to wafers
  – Wide range of impurity species (almost anything)
  – Tight dose control (A few % vs. 20--30% for high temperature pre--deposition processes)
  – Low temperature process
- Expensive systems
- Vacuum system

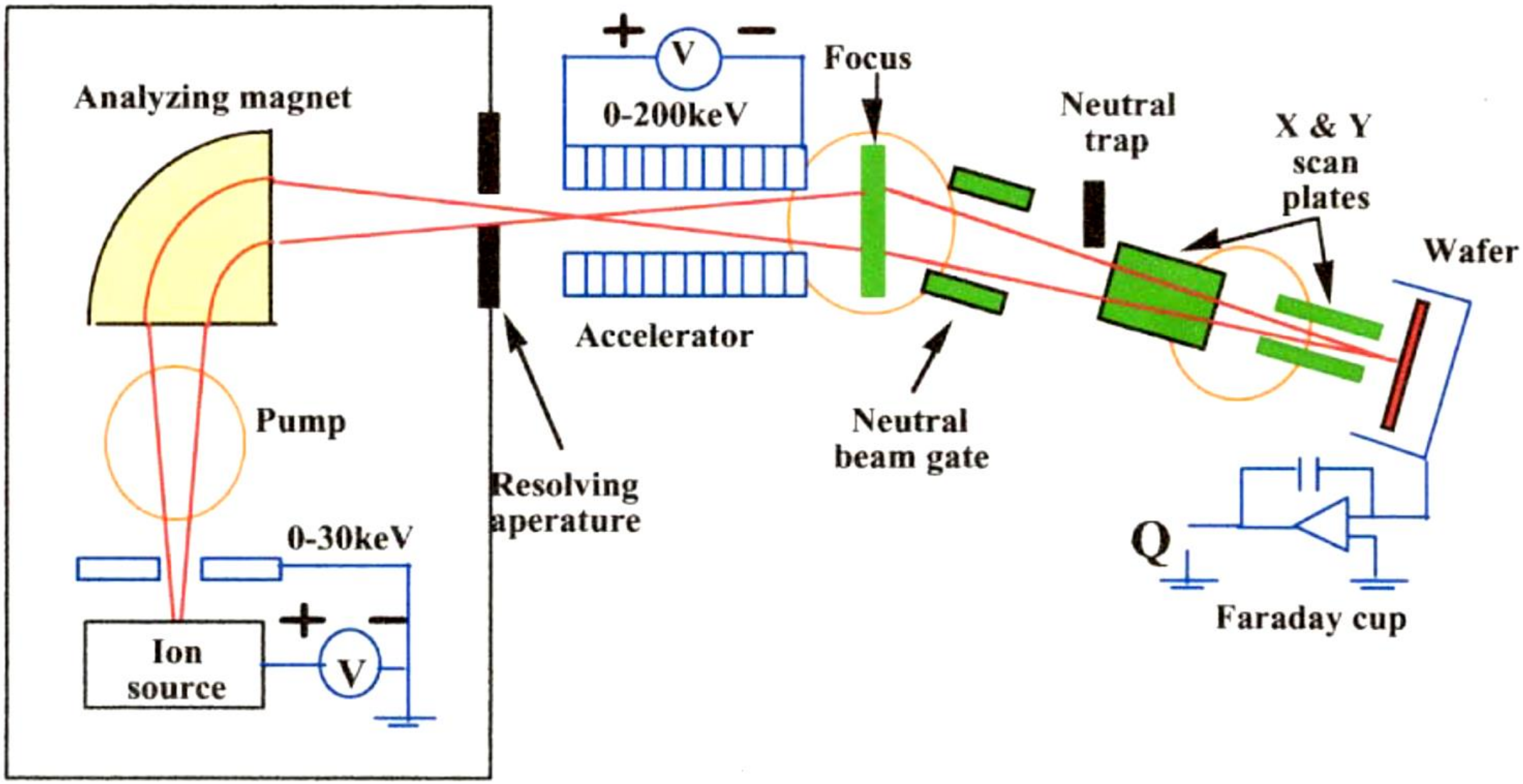
5.2.2 Schematics of ion implanter



5.2.2 ion implanter in AEMD

八英寸及以下，
3-960KeV,
B, BF2, P, A,Ar
元素的注入

5.2.2 Basics of ion implanter

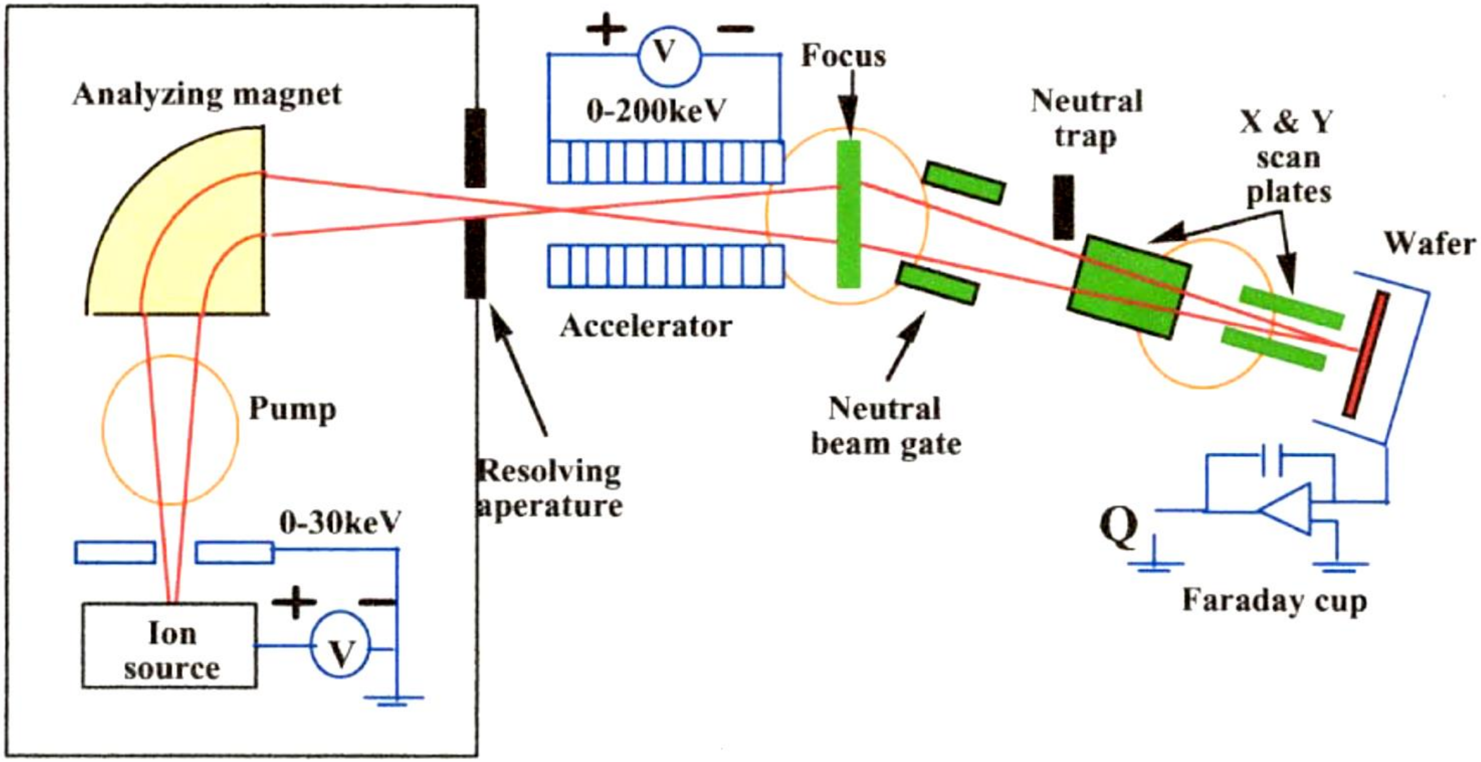


Ion source: BF_3, AsH_3, PH_3…

Mass analyzer: **m/z can be identified in magnetic field, thus, target ion beam can be extracted.**

Accelerator: **ion beam is accelerated under high voltage. It determine the depth of ion implantation.**

5.2.2 Basics of ion implanter



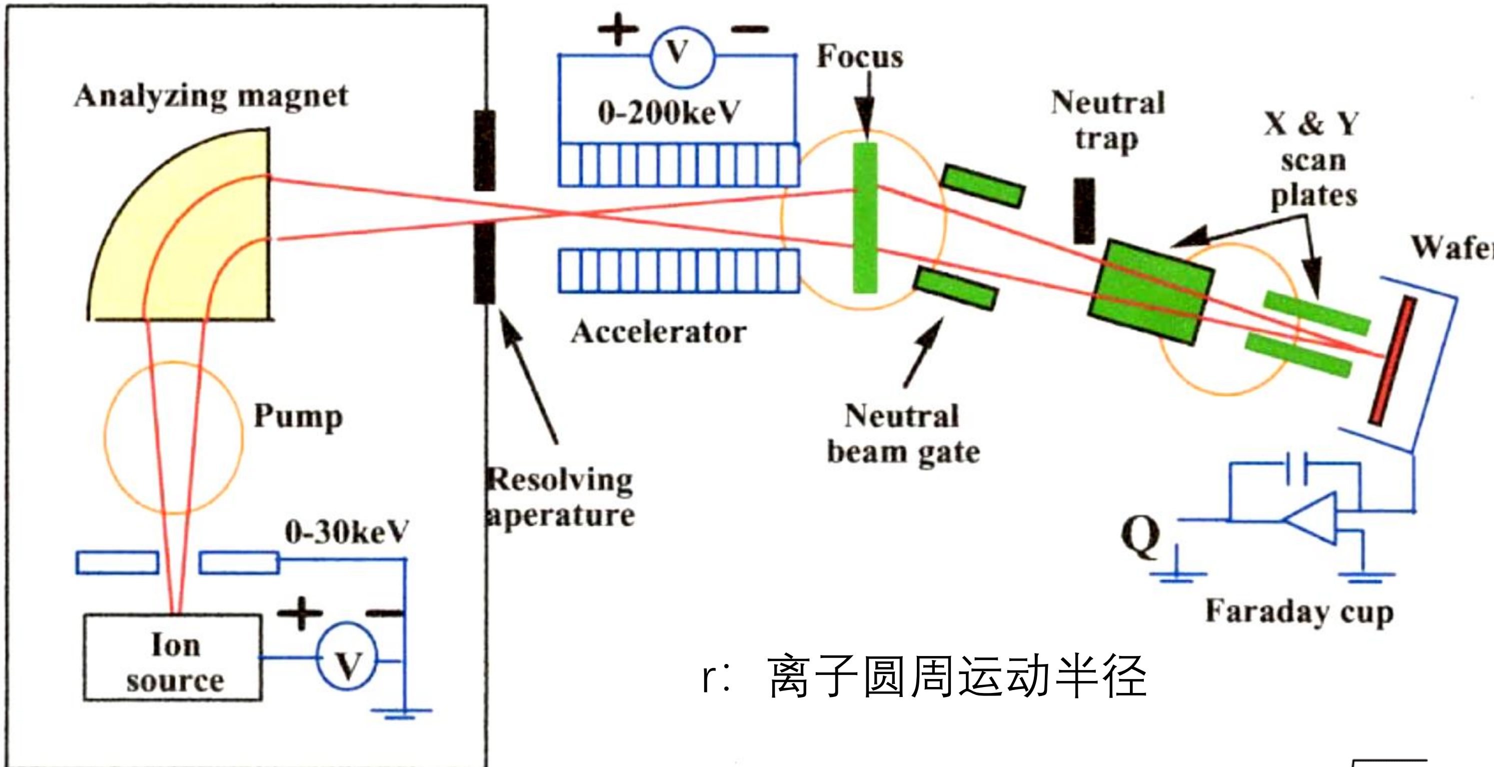
Neutral trapper: neutral atom is separated by deviation electrodes.

Focus system: focus ion beam to be mm in diameter.

Scan system: scan ion beam along x and y direction.

Operation chamber: set the wafer (samples). Position can be adjusted.

5.2.2 Basics of ion implanter



r：离子圆周运动半径

Force on charged particle

$$\frac{Mv^2}{r} = q(v \times B) \qquad v = \sqrt{\frac{2E}{M}} = \sqrt{\frac{2qV_{ext}}{M}}$$

Magnetic Field and current

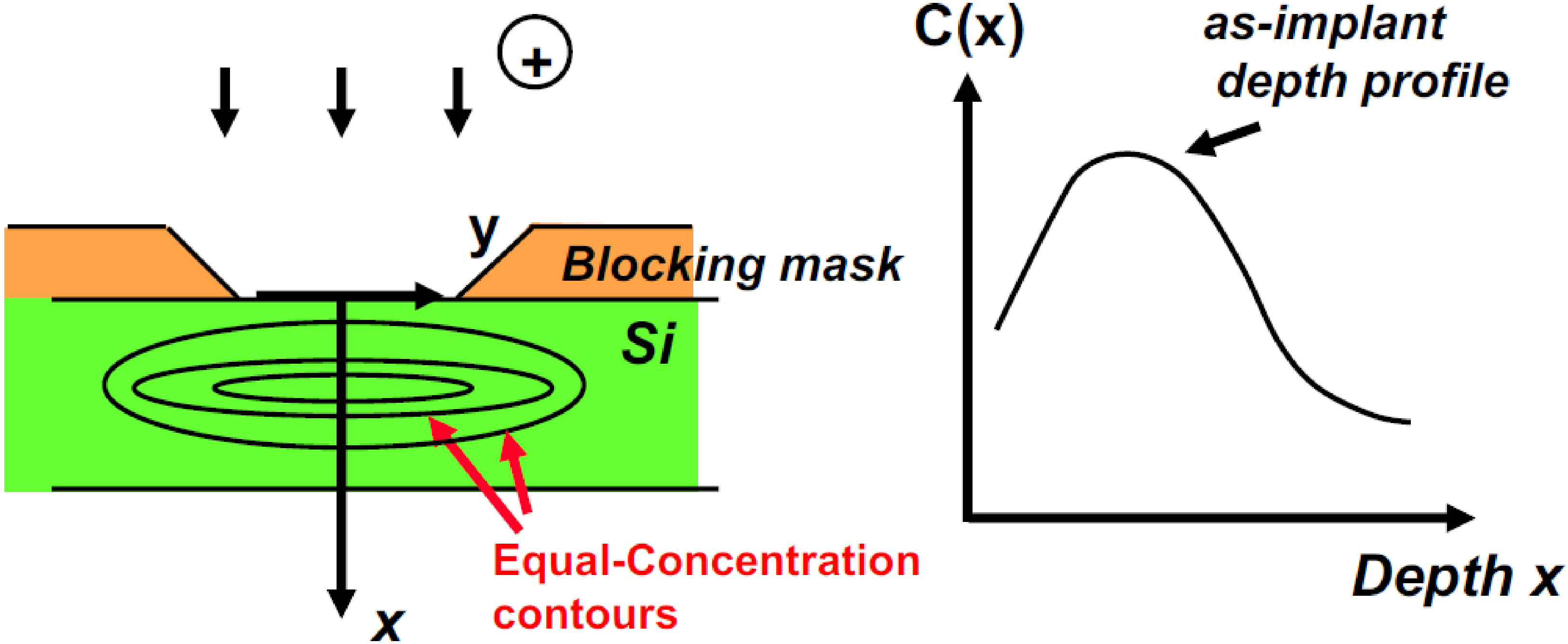
$$|B| = \alpha I = \sqrt{\frac{2mV_{ext}}{qr^2}}$$

Implanted Dose

$$Q = \frac{1}{nqA}\int_0^T I(t)dt$$

Outline

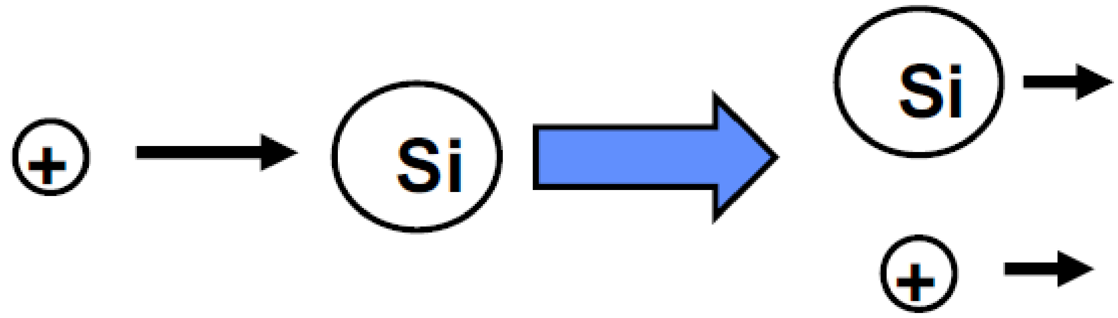
5.3.1 Basics of ion implantation



During implantation, temperature is ambient.
Post-implant annealing step (> 900°C) is required to anneal out defects.

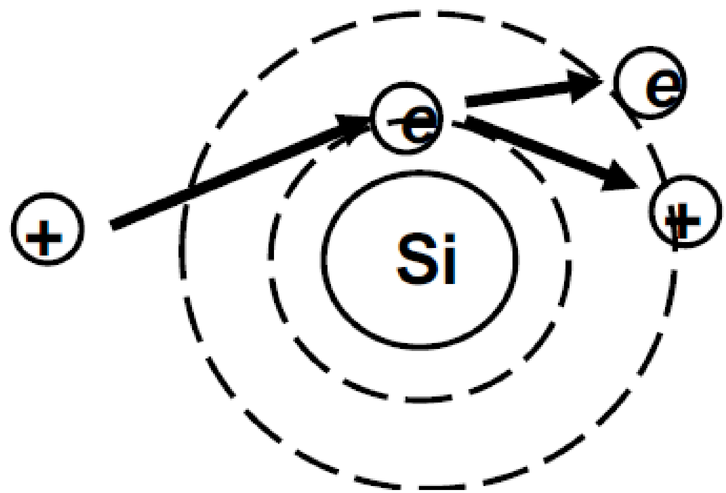
5.3.2 Ion Energy Loss Mechanisms

Nuclear stopping



Crystalline Si substrate damaged by collision

Electronic stopping



Electronic excitation creates heat

5.3.2 Ion Energy Loss Characteristics

Light ions/at higher energy → more electronic stopping

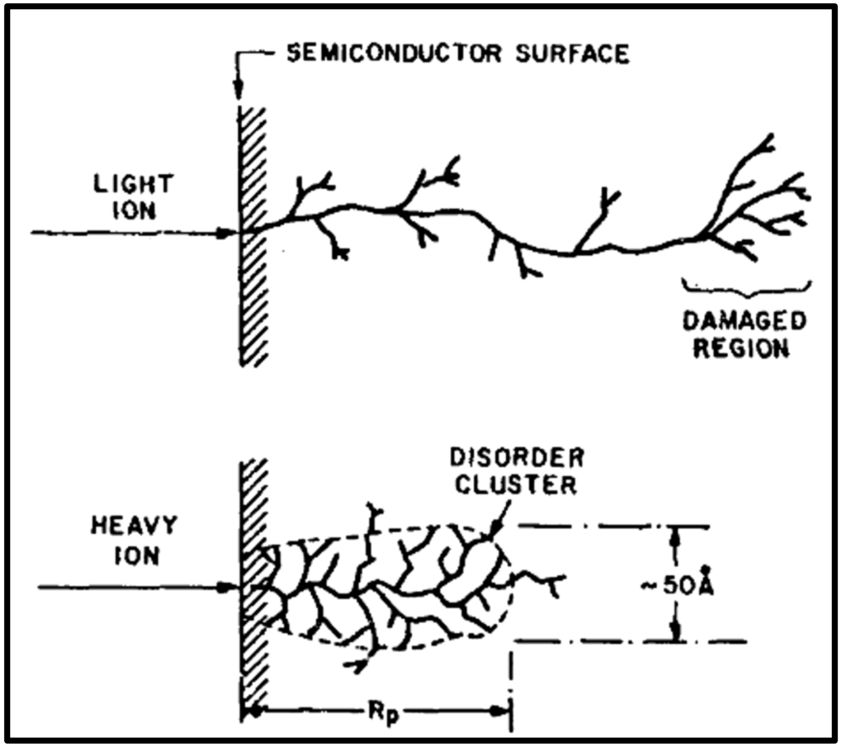
Heavier ions/at lower energy → more nuclear stopping

Examples: implanting into Si

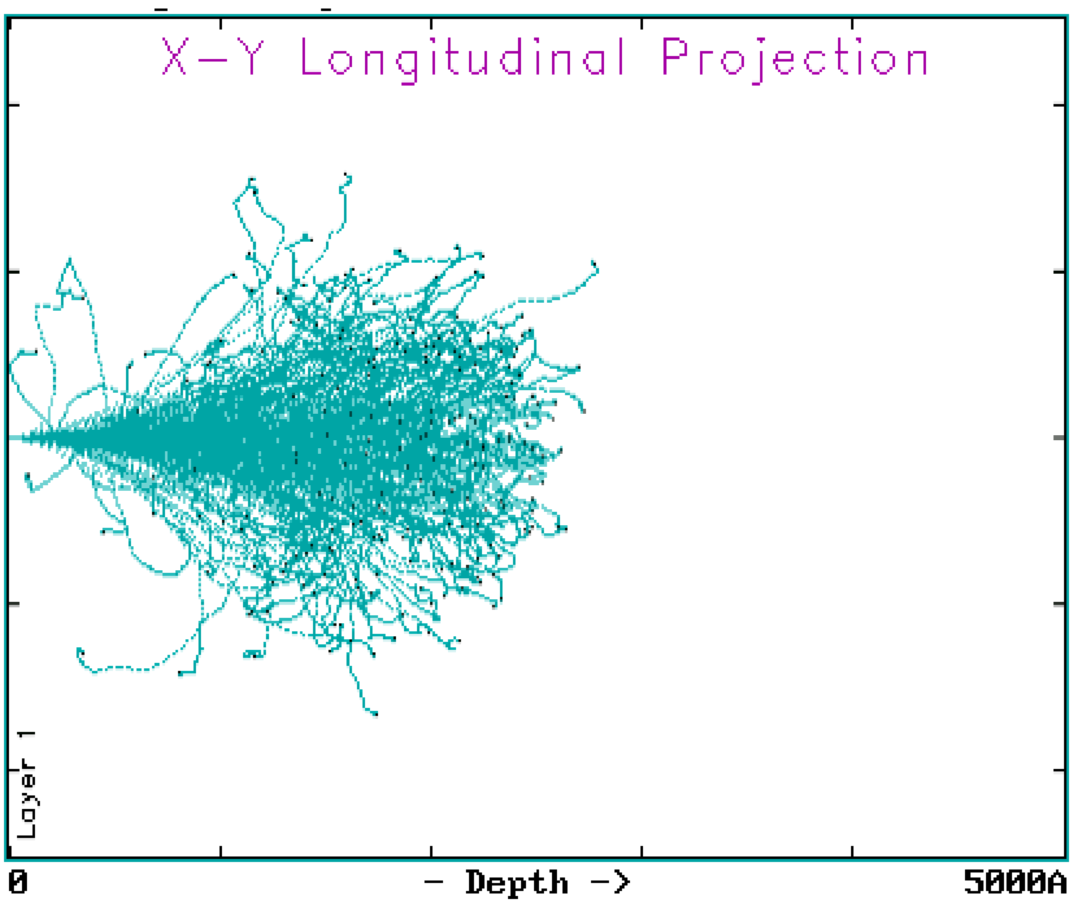
H^+ → **Electronic stopping dominates**

B^+ → **Electronic stopping dominates**

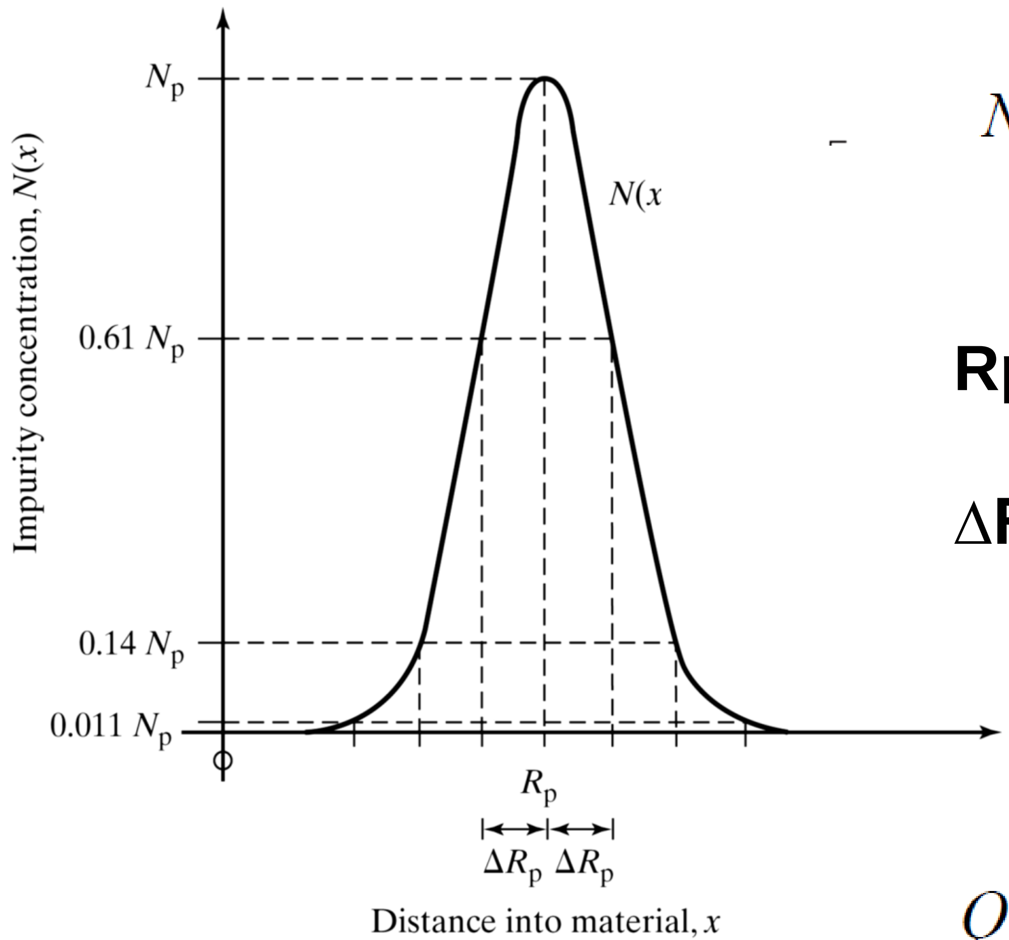
As^+ → **Nuclear stopping dominates**



Simulation of 50 keV Boron implanted into Si



5.3.3 Model for blanket implantation



$$N(x) = N_p \exp\left[-\frac{(x-R_p)^2}{2\Delta R_p^2}\right]$$

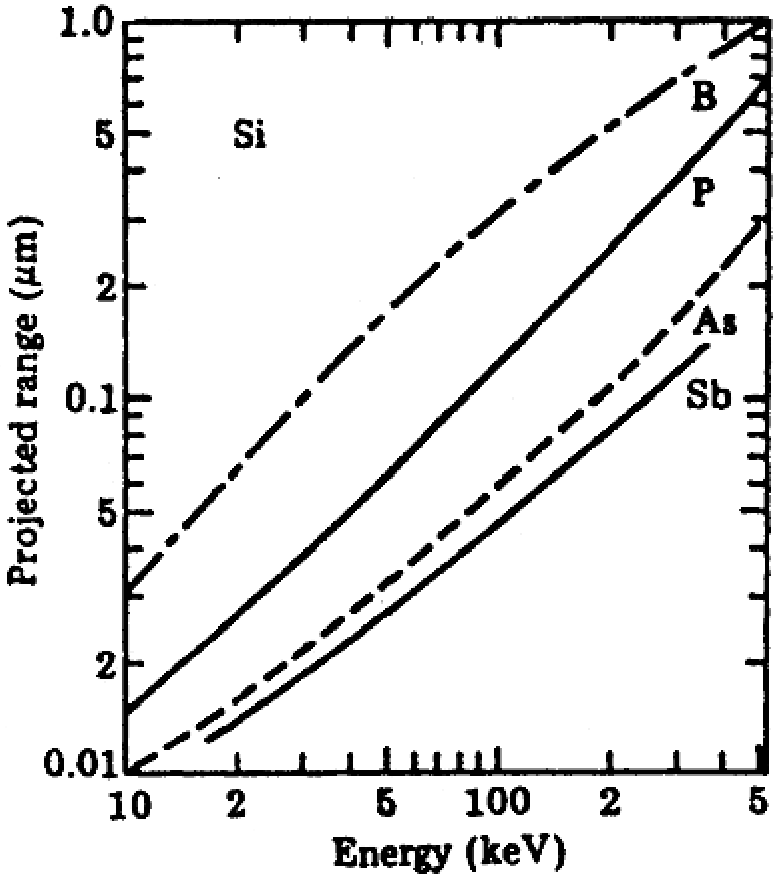
Rp: Projected Range (射程)

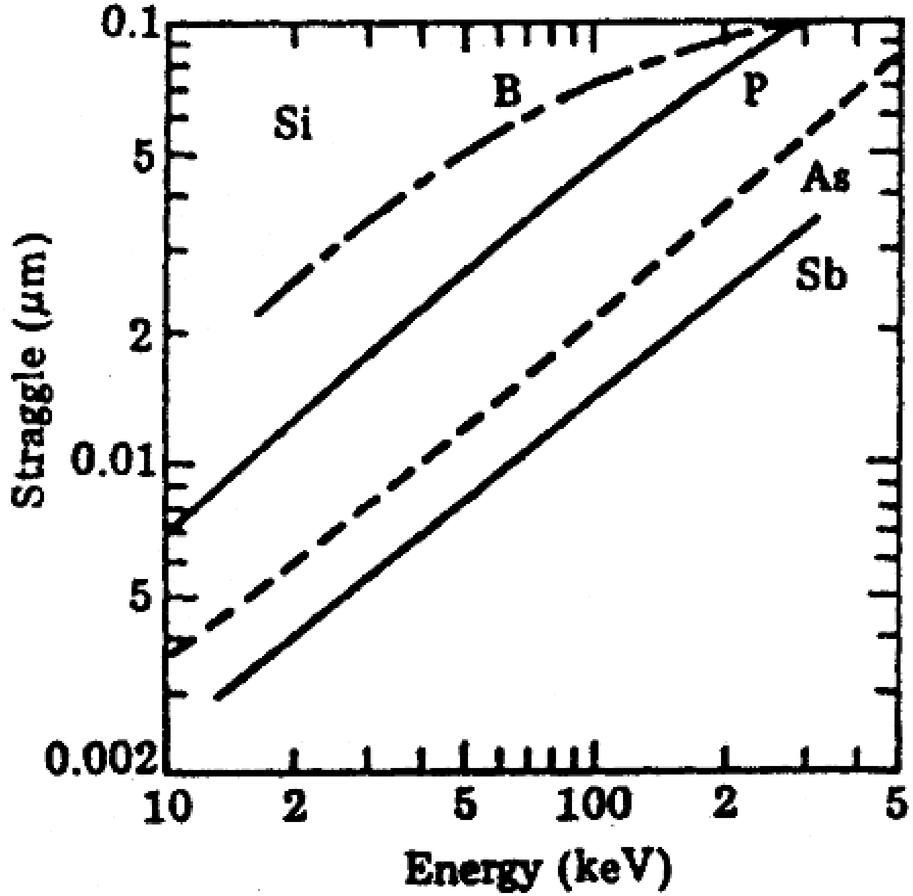
ΔRp: Straggle（标准偏差）

$$Q = \int_0^{\infty} N(x)dx = \sqrt{2\pi} N_p \Delta R_p$$

5.2.3 Projected Range and Straggle

R_p and ΔR_p values are given in tables or charts





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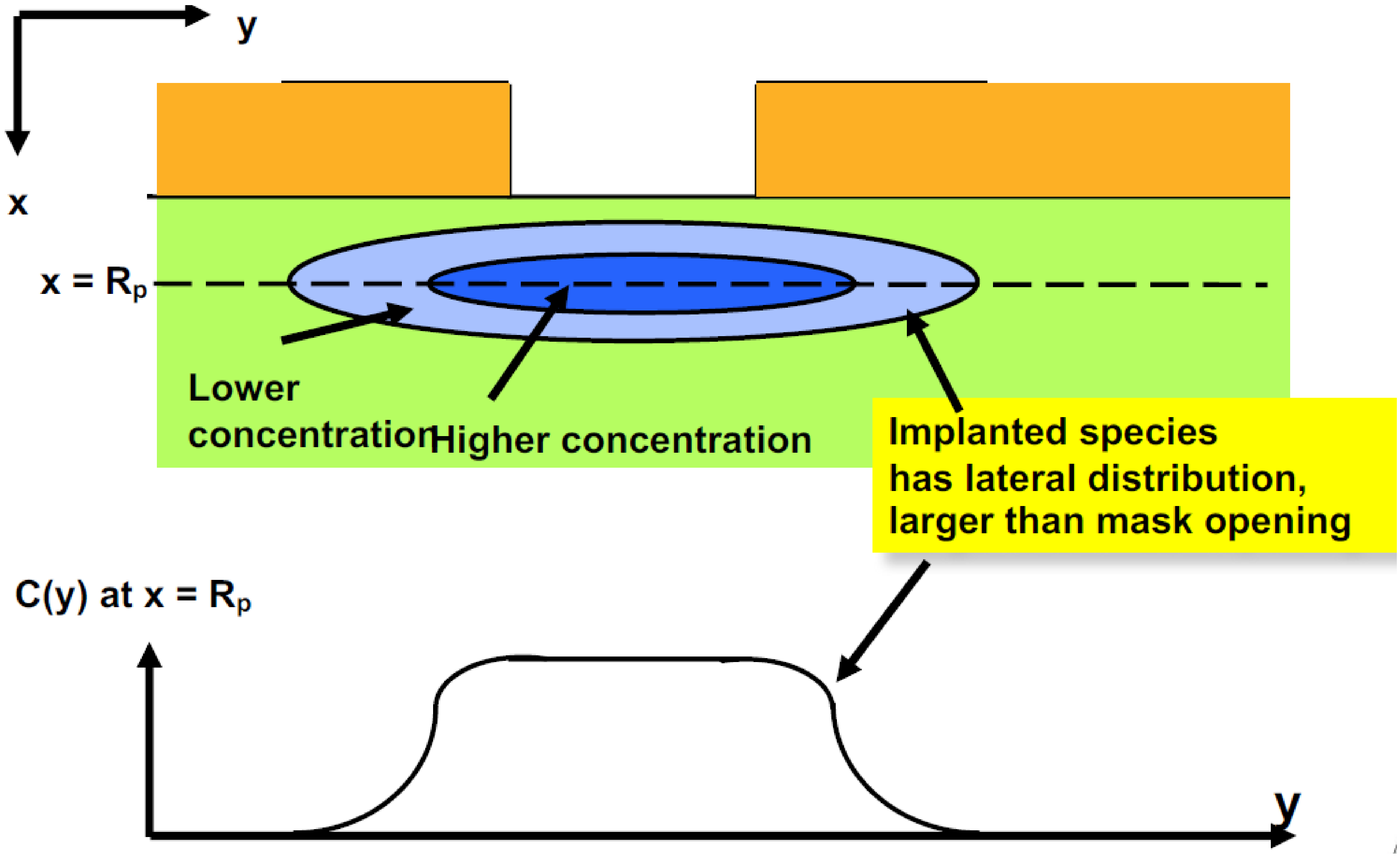
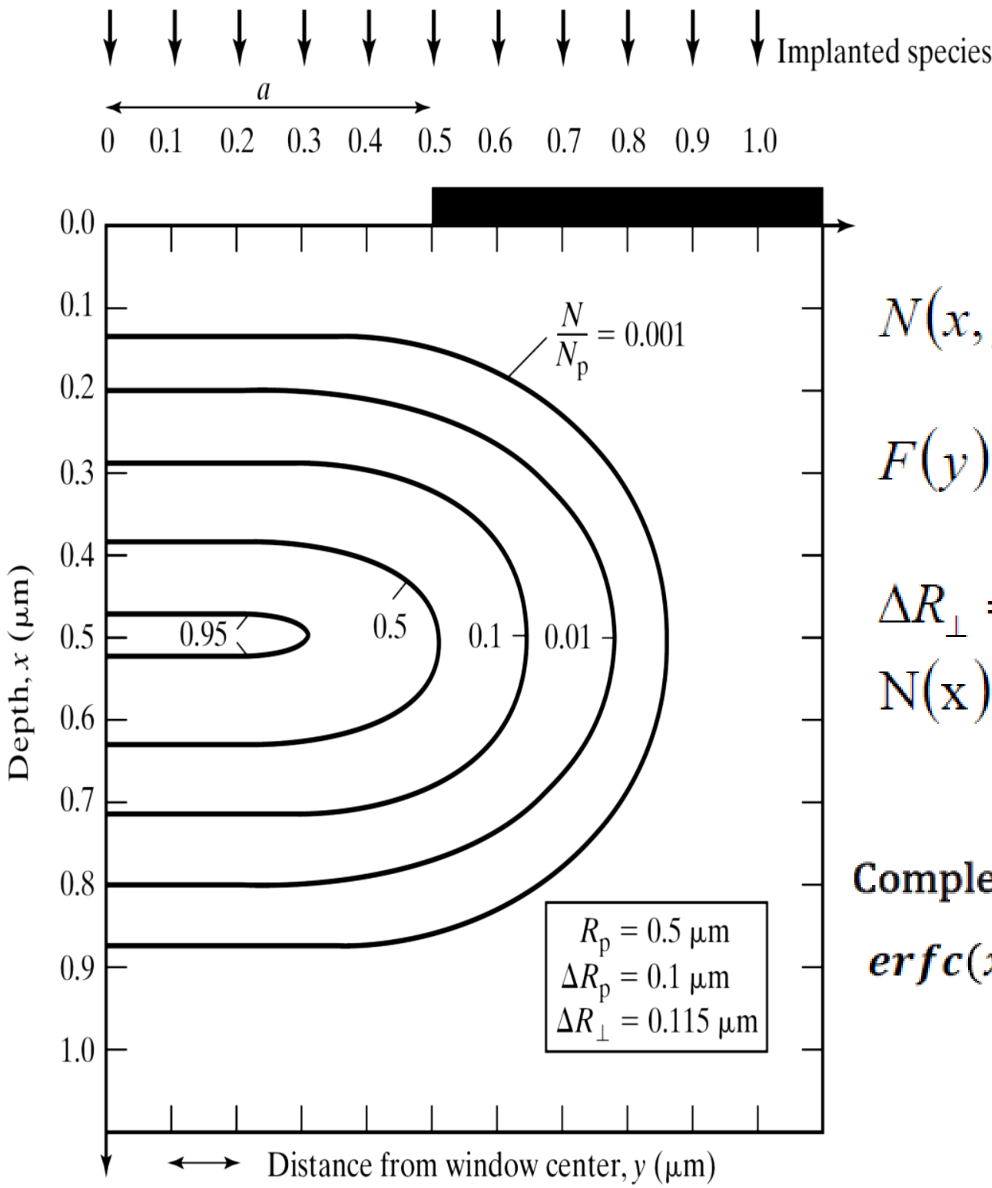
5.4.1 Selective ion implantation

Feature Enlargement due to Lateral Straggle



5.4.1 Selective ion implantation



$$N(x,y) = N(x)F(y)$$

$$F(y) = \frac{1}{2}\left[erfc\left(\frac{y-a}{\sqrt{2}\Delta R_\perp}\right) - erfc\left(\frac{y+a}{\sqrt{2}\Delta R_\perp}\right)\right]$$

$\Delta R_\perp$ = transverse straggle 横向标准偏差

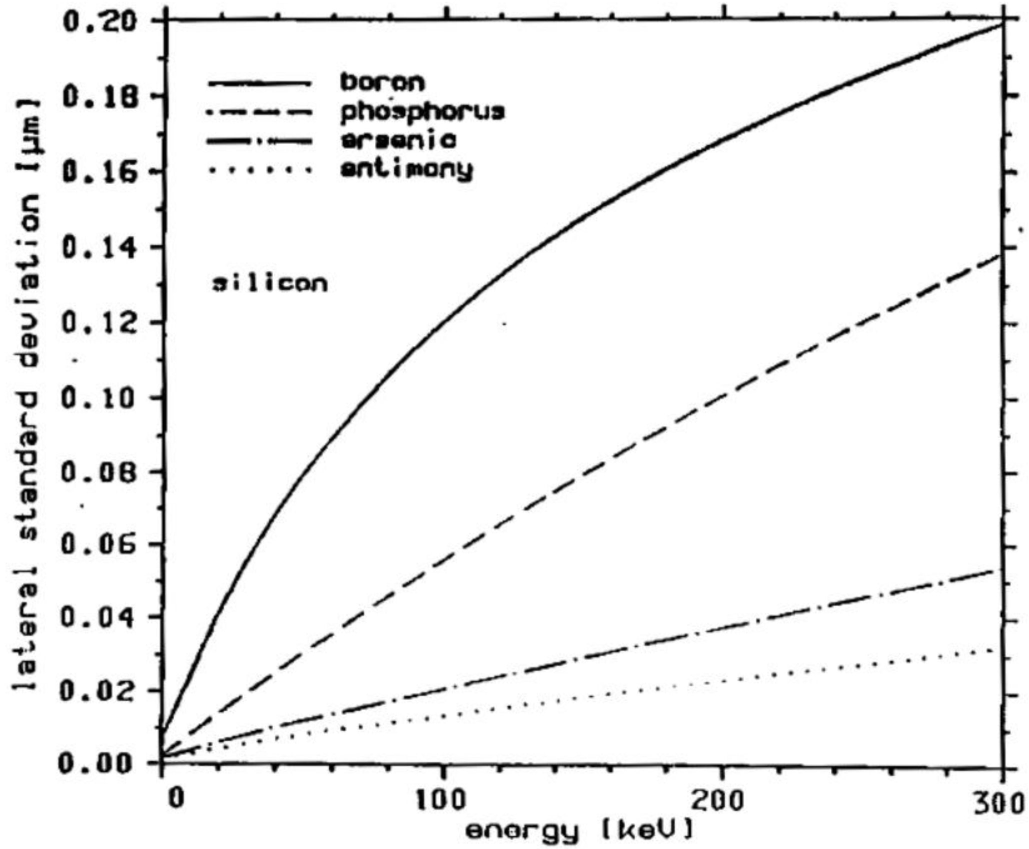
N(x) is one-dimensional solution

Complementary error function:

$$erfc(x) = 1 - erf(x) = \frac{2}{\sqrt{\pi}}\int_x^\infty e^{-t^2}dt$$

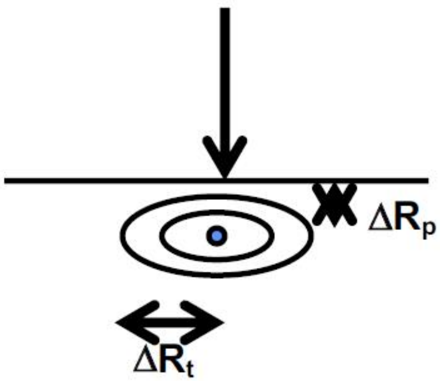
5.4.1 Transversal straggle

ΔR_t

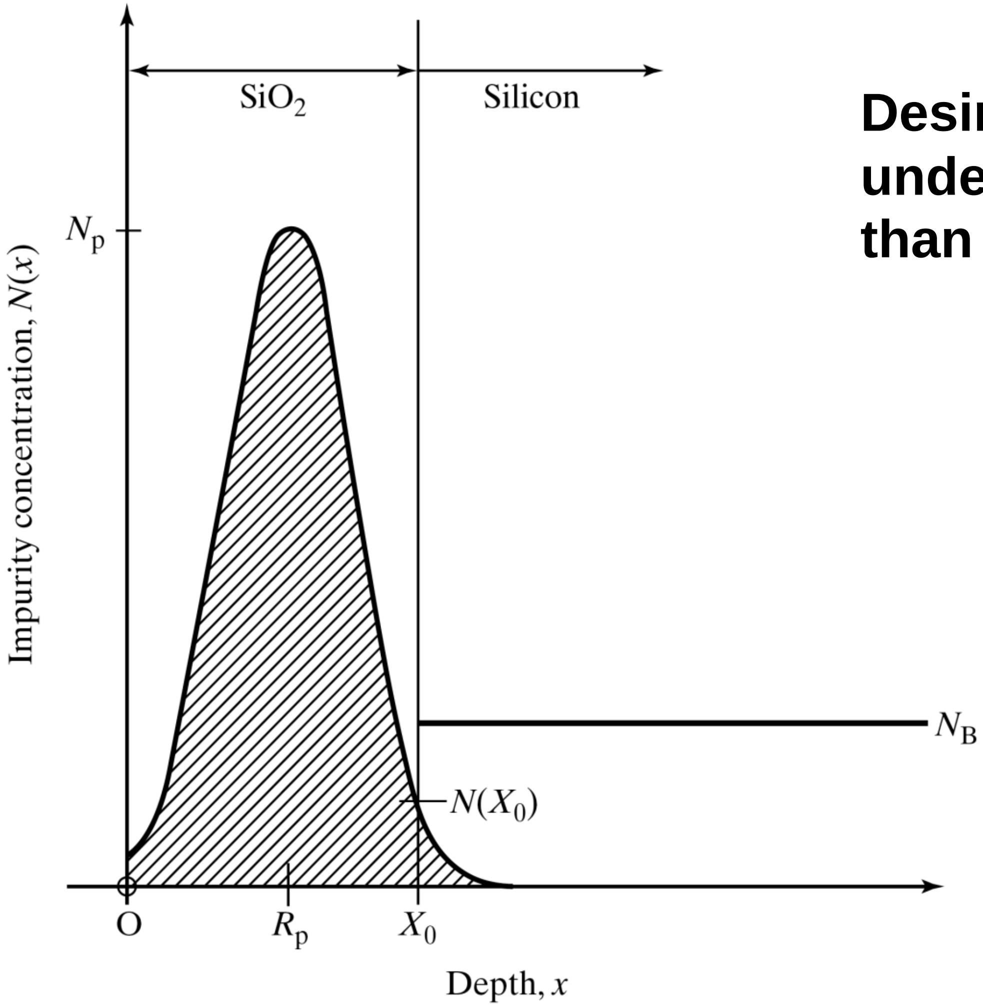


Lateral standard deviation of boron, phosphorus, arsenic and antimony in silicon

$$\frac{\Delta R_t}{\Delta R_\perp} > 1$$



5.4.2 Mask thickness



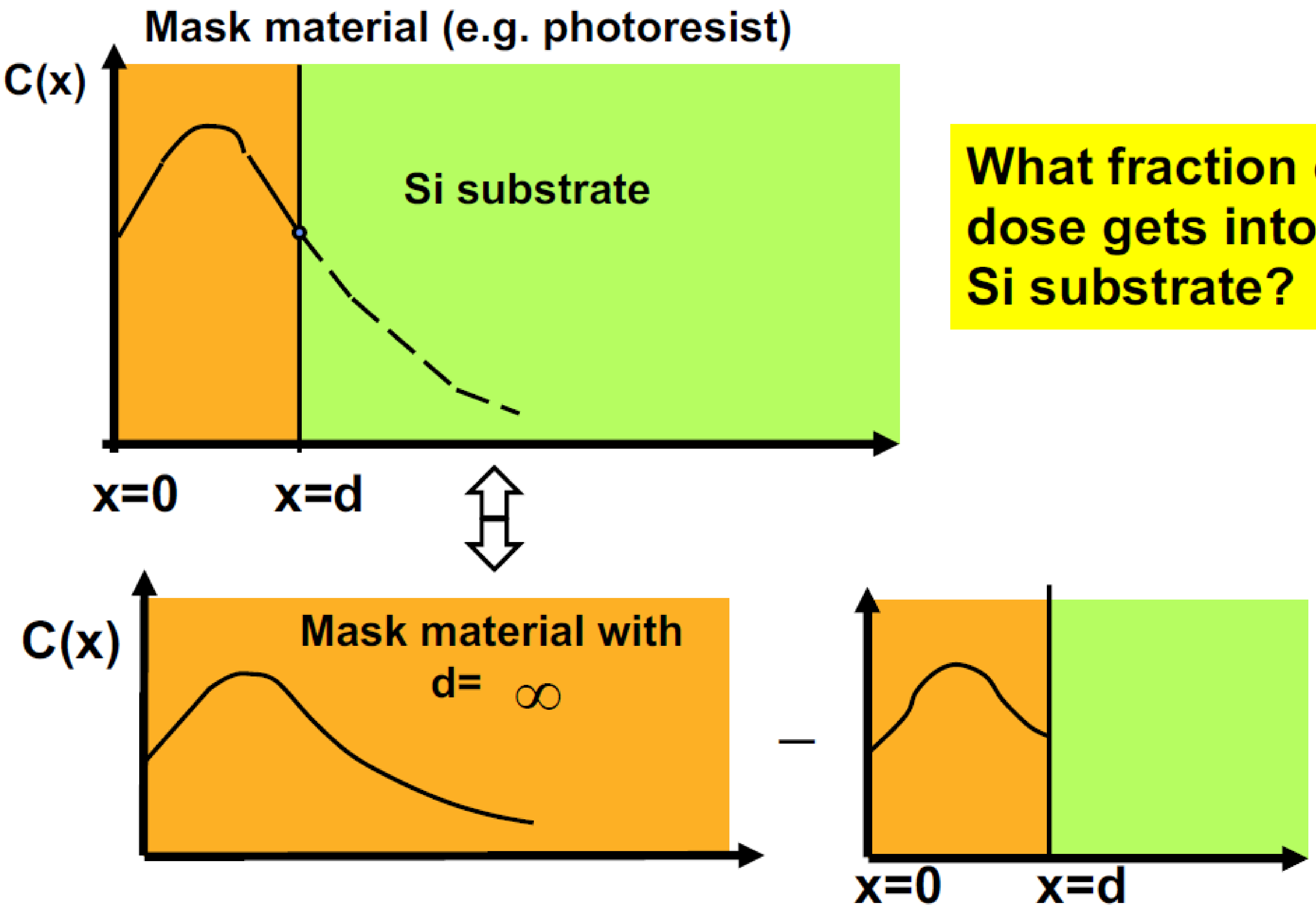
Desire implanted impurity level under the mask should be much less than background doping

$$N(x_0) \ll N_B$$

or

$$N(x_0) < \frac{N_B}{10}$$

5.4.2 Transmission factor of mask



What fraction of dose gets into Si substrate?

$$T = \int_0^{\infty} C(x)dx - \int_0^{d} C(x)dx$$

$$= \frac{1}{2} erfc\left\{\frac{d - R_p}{\sqrt{2}\Delta R_p}\right\}$$

R_p, ΔR_p are values of for ions into the masking material

$$erfc(x) = 1 - \frac{2}{\sqrt{\pi}} \int_0^{x} e^{-y^2} dy$$

Rule of thumb: Good masking thickness

$$d = R_p + 4.3\Delta R_p \qquad \frac{C(x = d)}{C(x = R_p)} \sim| 10^{-4}$$

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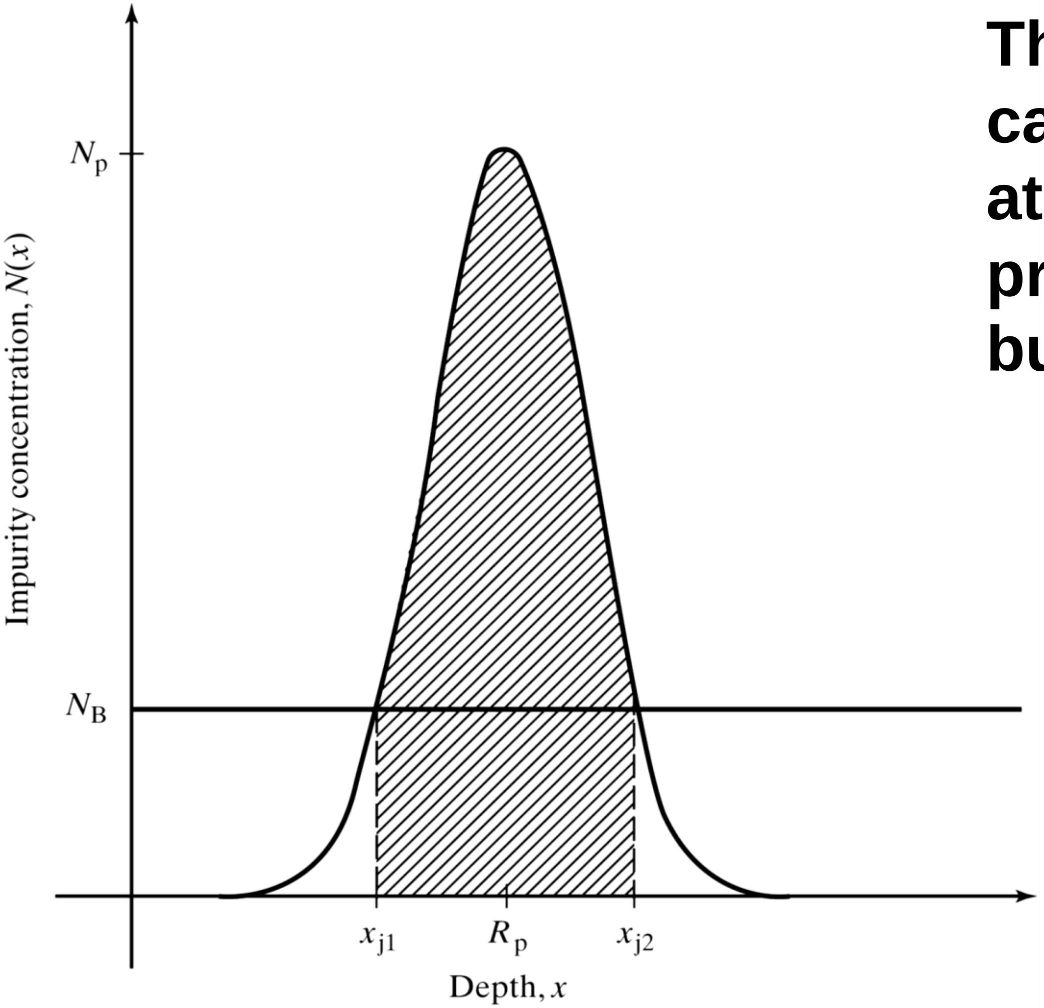
5.5.1 Junction depth



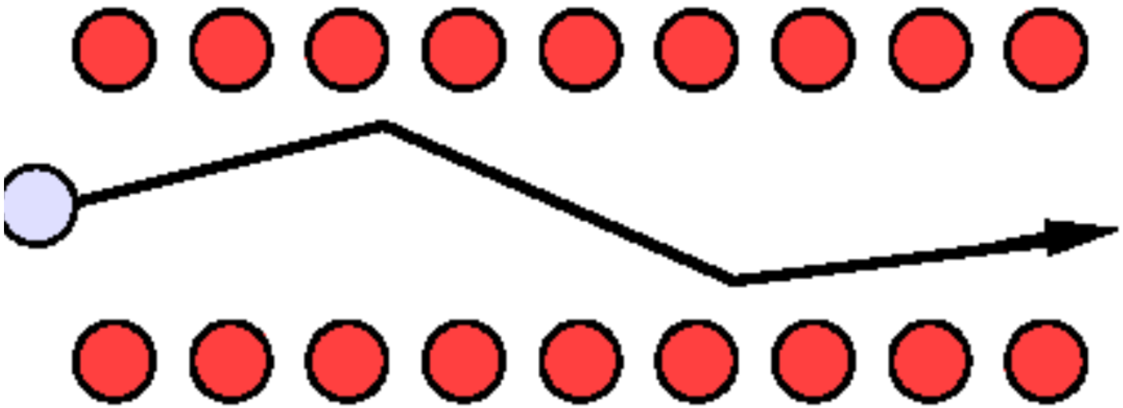
The junction depth is calculated from the point at which the implant profile concentration = bulk concentration:

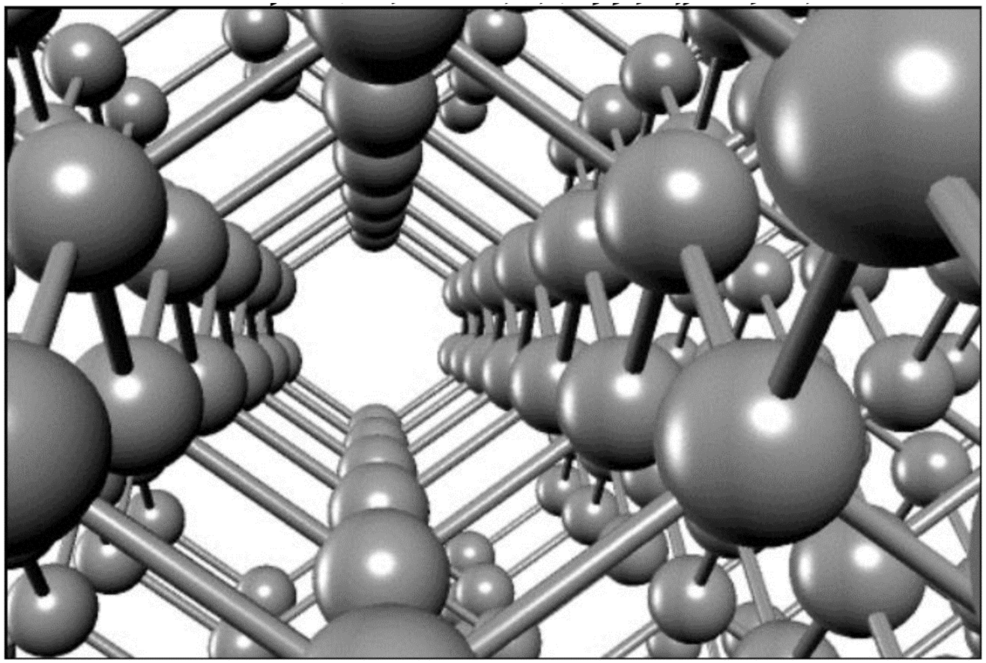
$$N(x_j) = N_B$$

$$N_p \exp\left[-\frac{(x_j - R_p)^2}{2\Delta R_p^2}\right] = N_B$$

$$x_j = R_p \pm \Delta R_p \sqrt{2\ln\left(\frac{N_p}{N_B}\right)}$$

5.5.2 Channeling

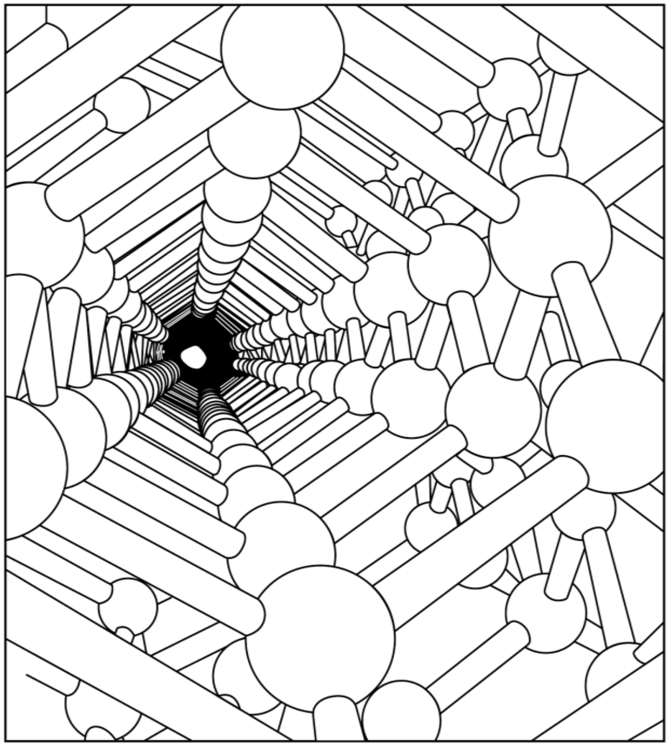




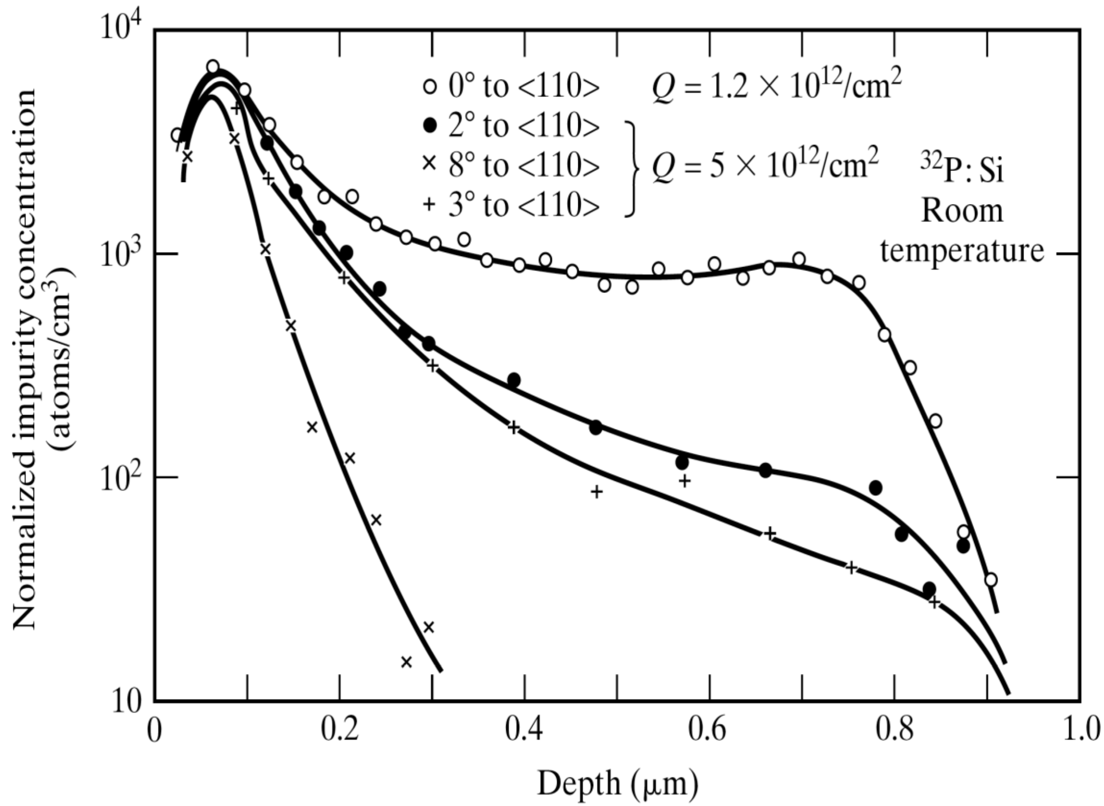
When ions is implanted along the direction of crystal axis, they cannot be scattered by the nucleus. So, the direction of the ions is not change so much that they can go ahead farther.

5.5.2 Channeling

Along (110)



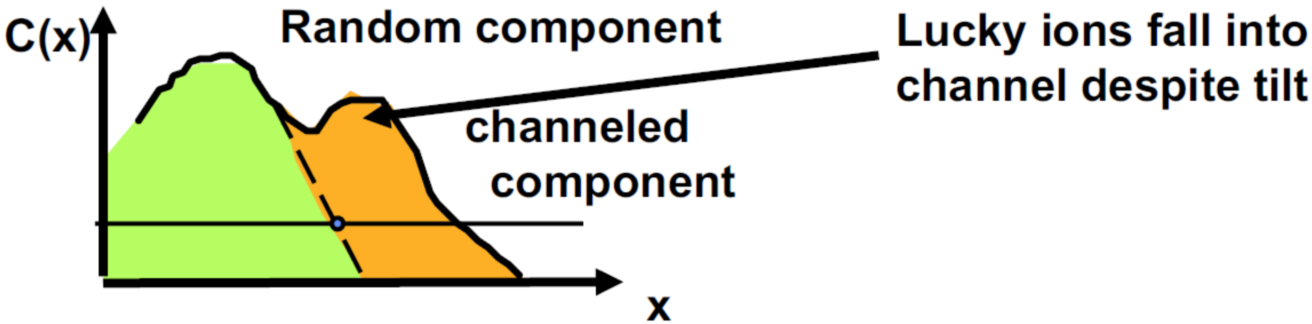
Effect of channeling



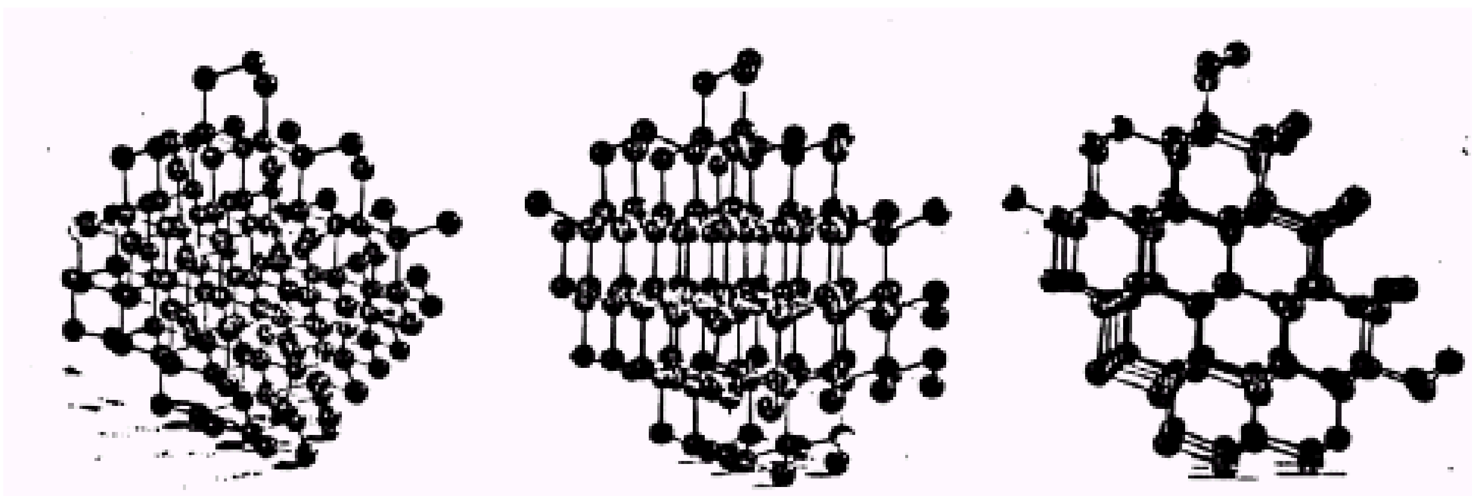
Due to channeling effect, it is hard to control the depth of implantation.

5.5.2 method to reduce channeling (1)

Use of tilt to reduce channeling



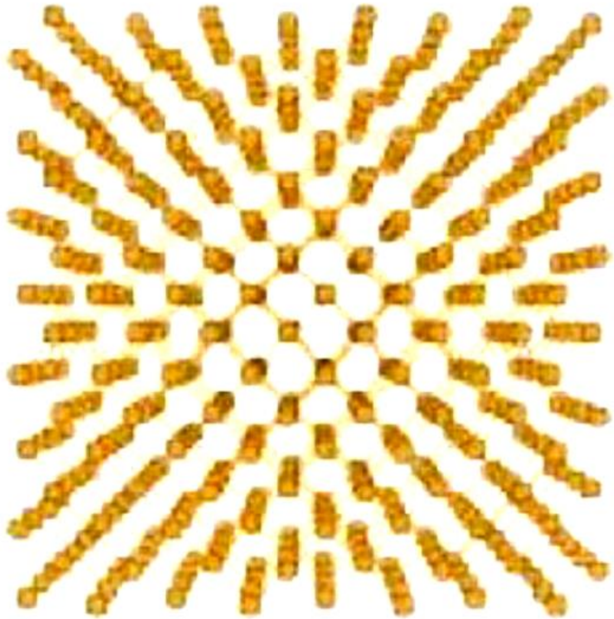
Random | Planar Channeling | Axial Channeling



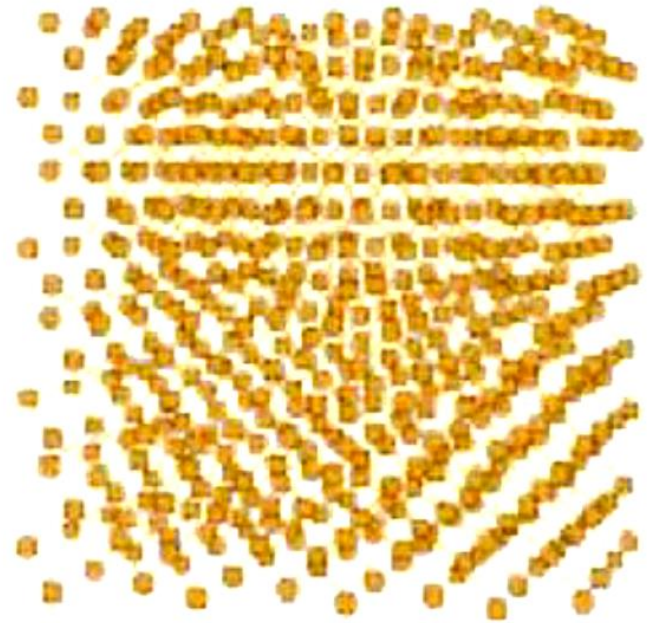
To minimize channeling, we tilt wafer by 7° with respect to ion beam.

5.5.2 method to reduce channeling (1)

Looking along <100> axis



With a 7° tilt and a rotation to simulate a “random” direction.

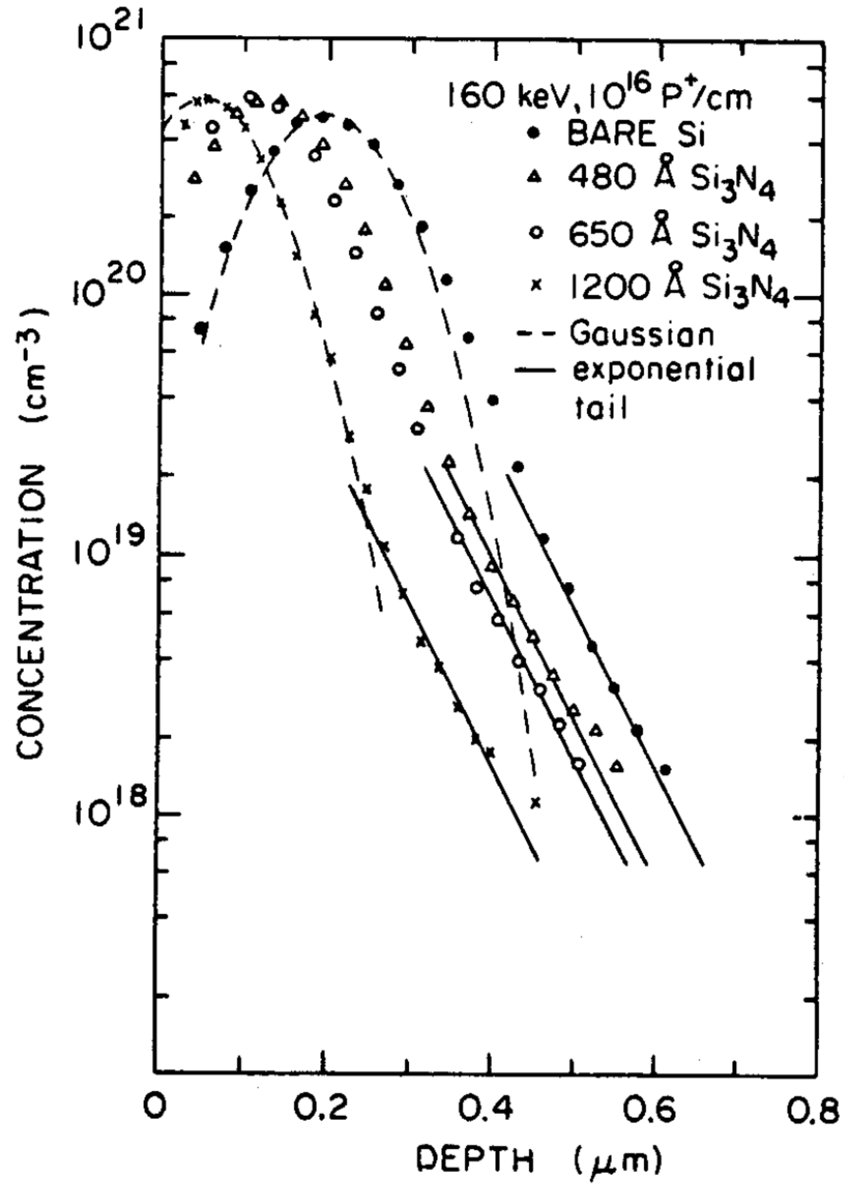


Tilting and rotating can be used to minimize channeling, but channeling can never be totally eliminated, because:

- **incident ions can be scattered into channeling directions in their first collisions**
- **ions that are not channeled early in the process have an increasing probability to be scattered into channels as they proceed because their energy is reduced and the critical angle increases**

5.5.2 method to reduce channeling (2)

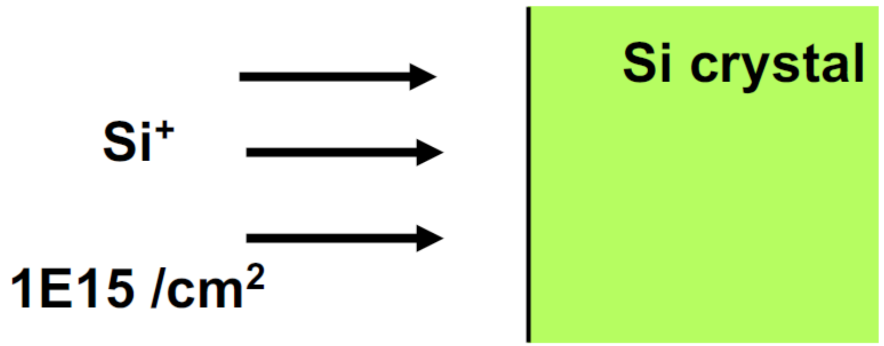
Amorphous layer is used



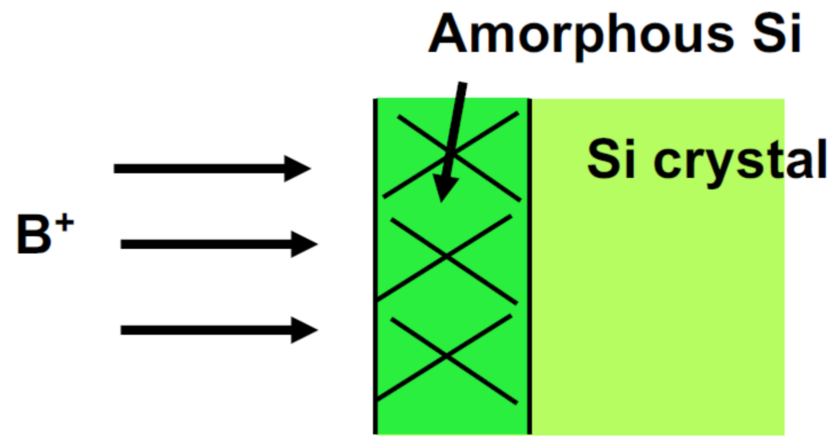
all,

Prevention of Channeling by Pre-amorphization

Step 1
High dose Si+ implantation to covert surface layer into amorphous Si

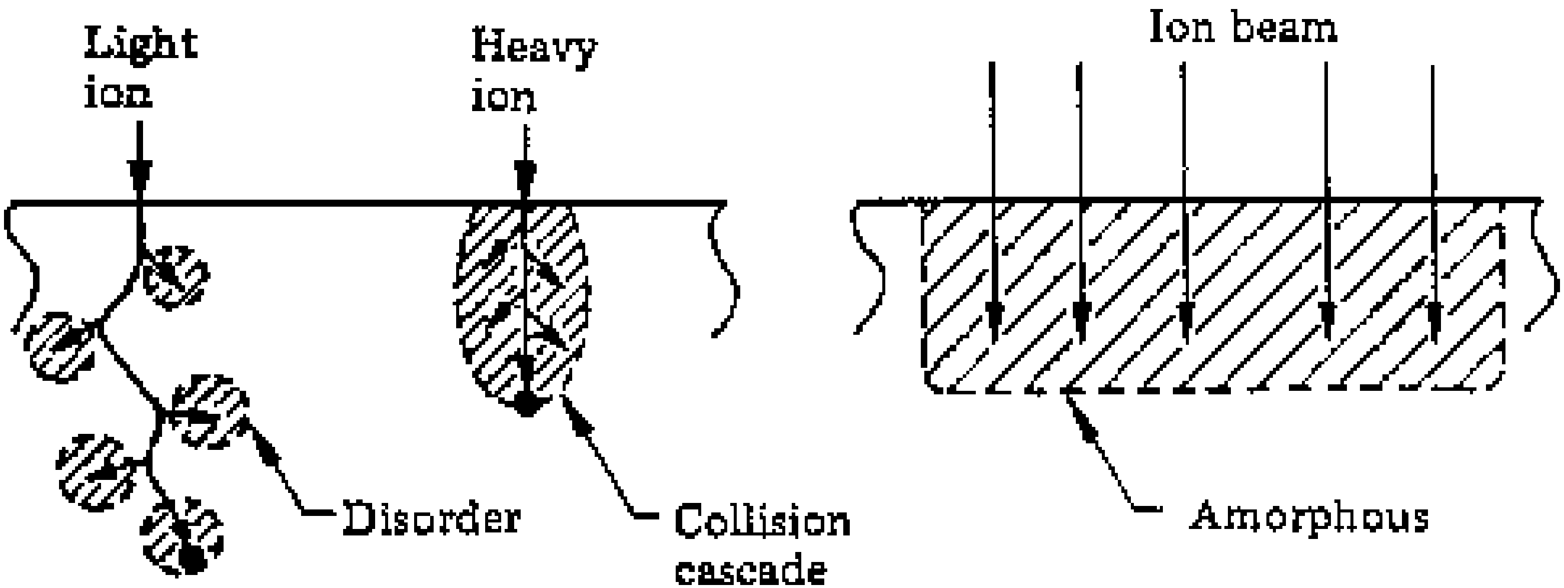


Step 2
Implantation of desired dopant into amorphous surface layer



Disadvantage:
Needs an additional high—dose implantation step

5.5.3 Implantation damage



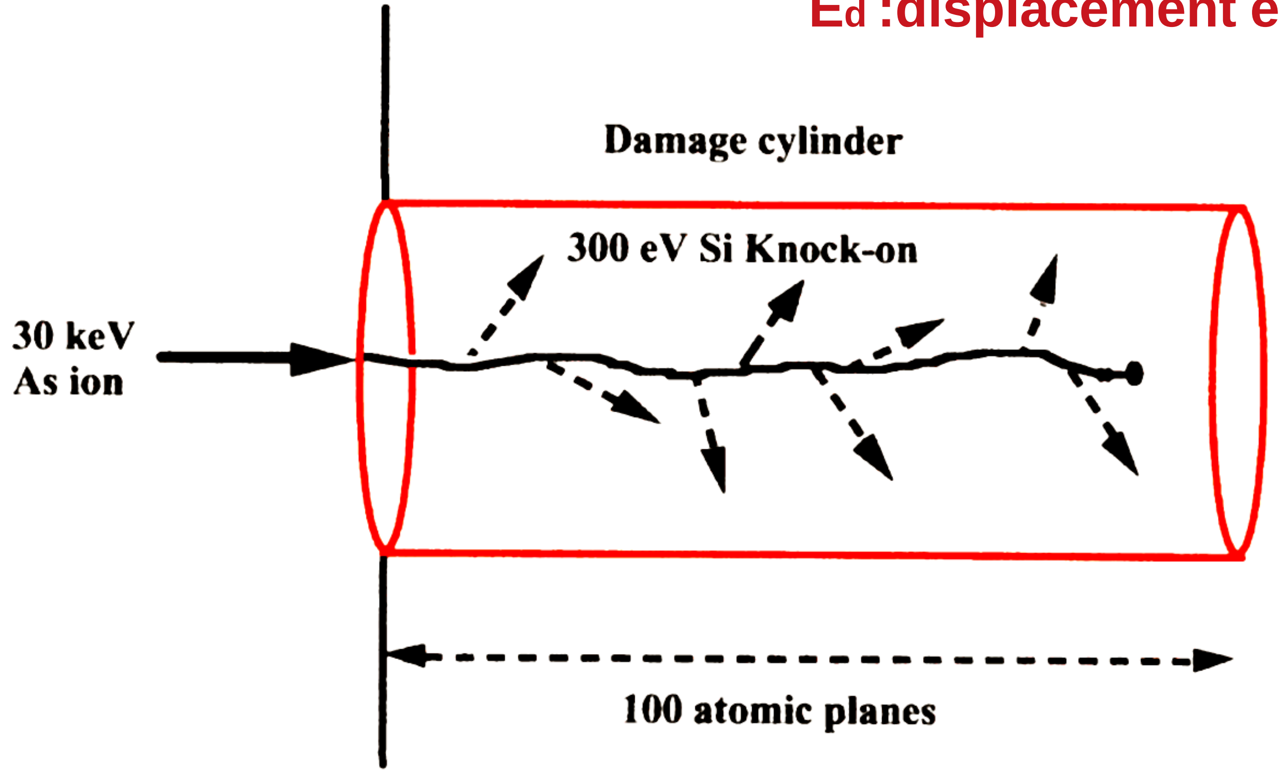
Dislocation and collision cascade are produced in Si crystal in ion Implantation causing Si amorphization with high enough does

5.5.3 Implantation damage

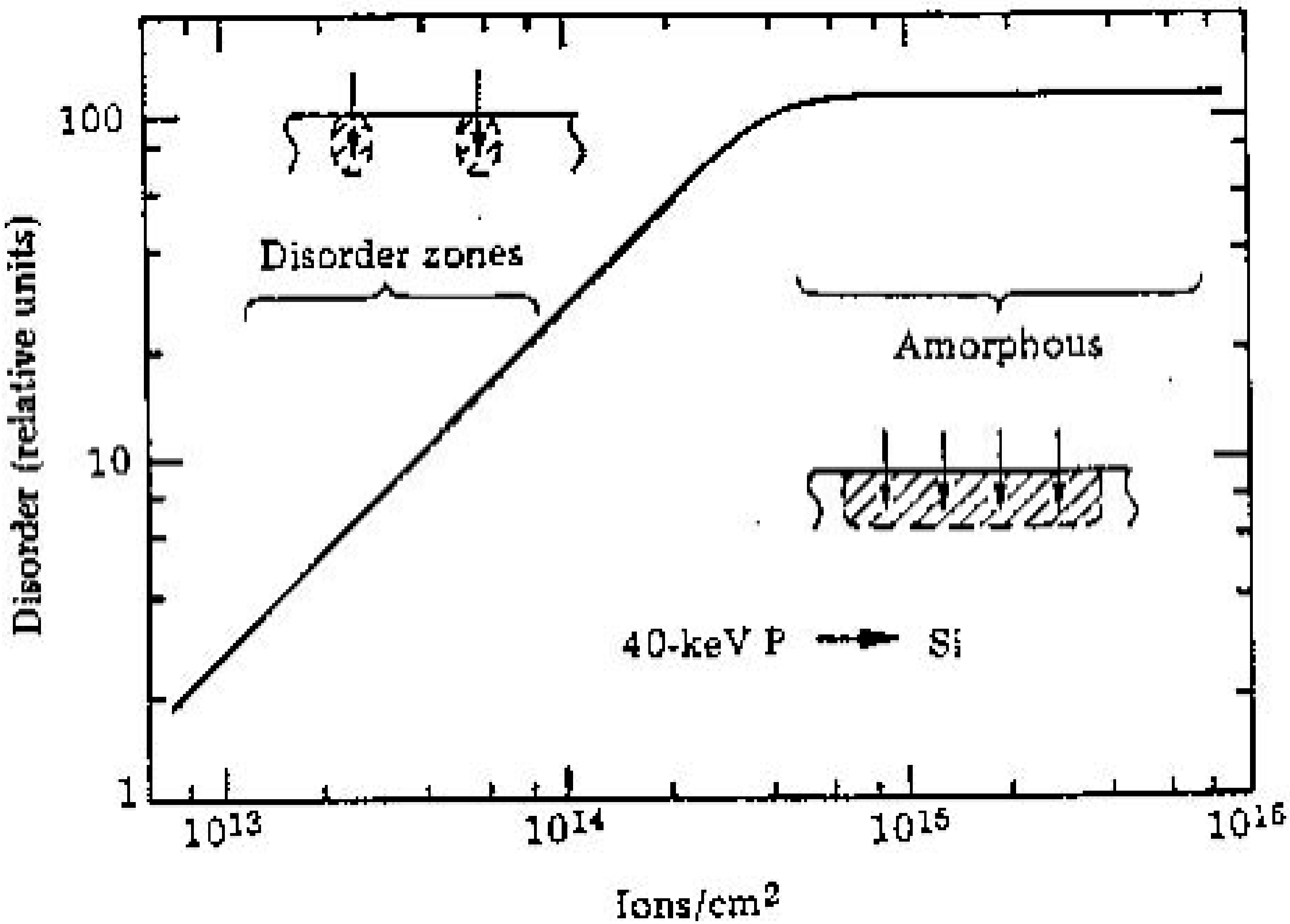
- The number of displaced particles created by an incoming ion is given by

$$n = \frac{E_n}{2E_d} = \frac{30,000}{2 \times 15} = 1000 \quad \text{ions} \qquad (17)$$

E_d :displacement energy

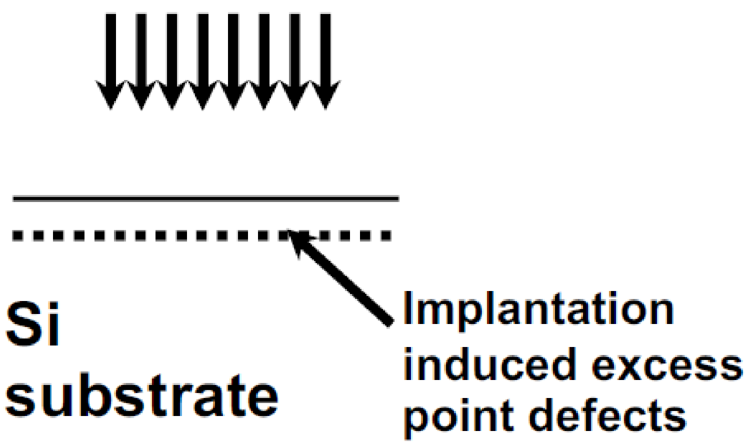


Amount and Type of Crystalline Damage

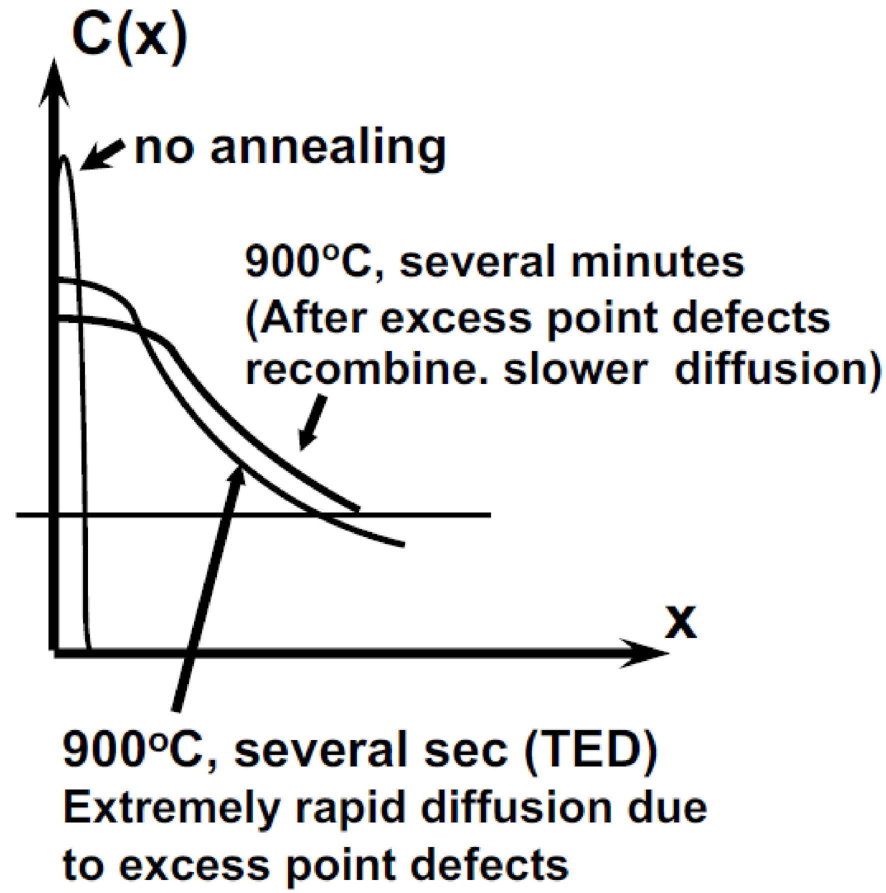


5.5.3 Transient Enhanced Diffusion

Dopant Implantation



Implantation creates large number of excess Si interstitials and vacancies. (1000X than thermal process). After several seconds of annealing, the excess point defects recombine.



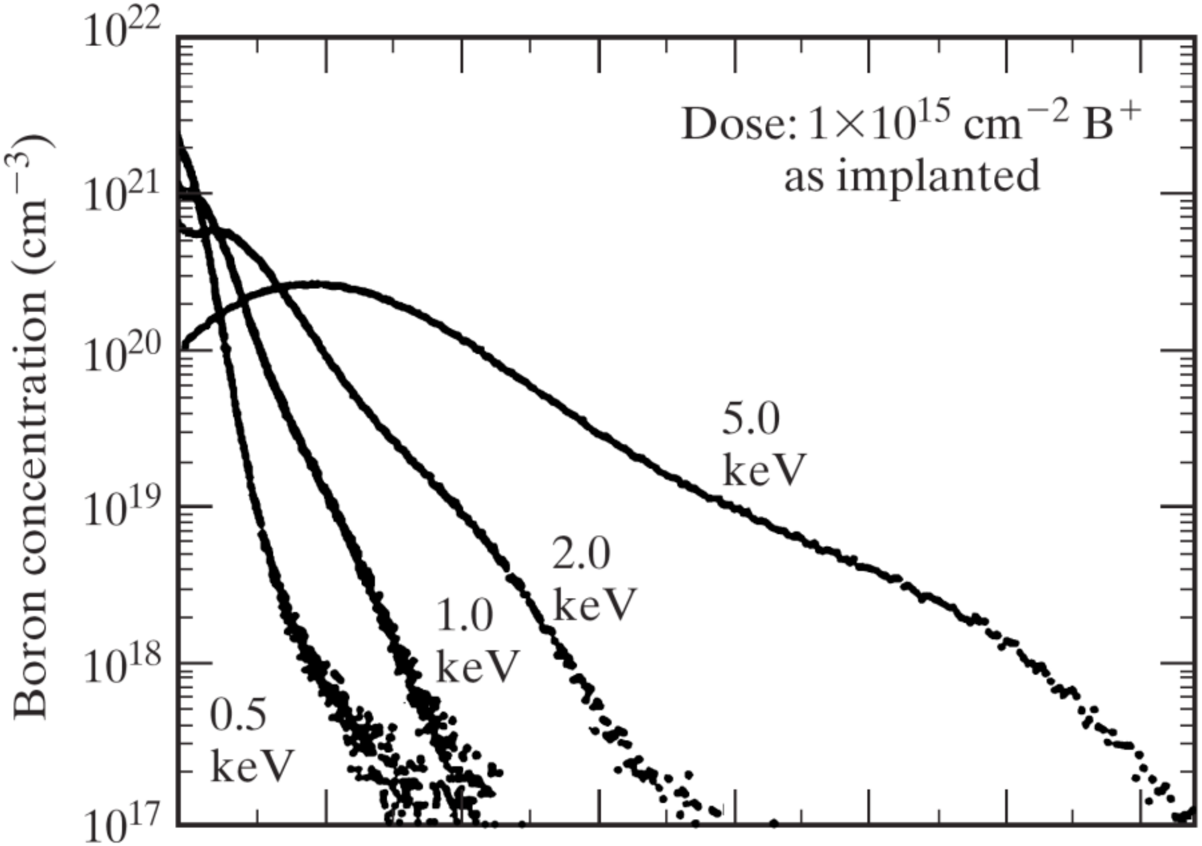
Post--Implantation Annealing Summary

- **After implantation, we need an annealing step**
- **A typical anneal will:**

1. Restore Si crystallinity.

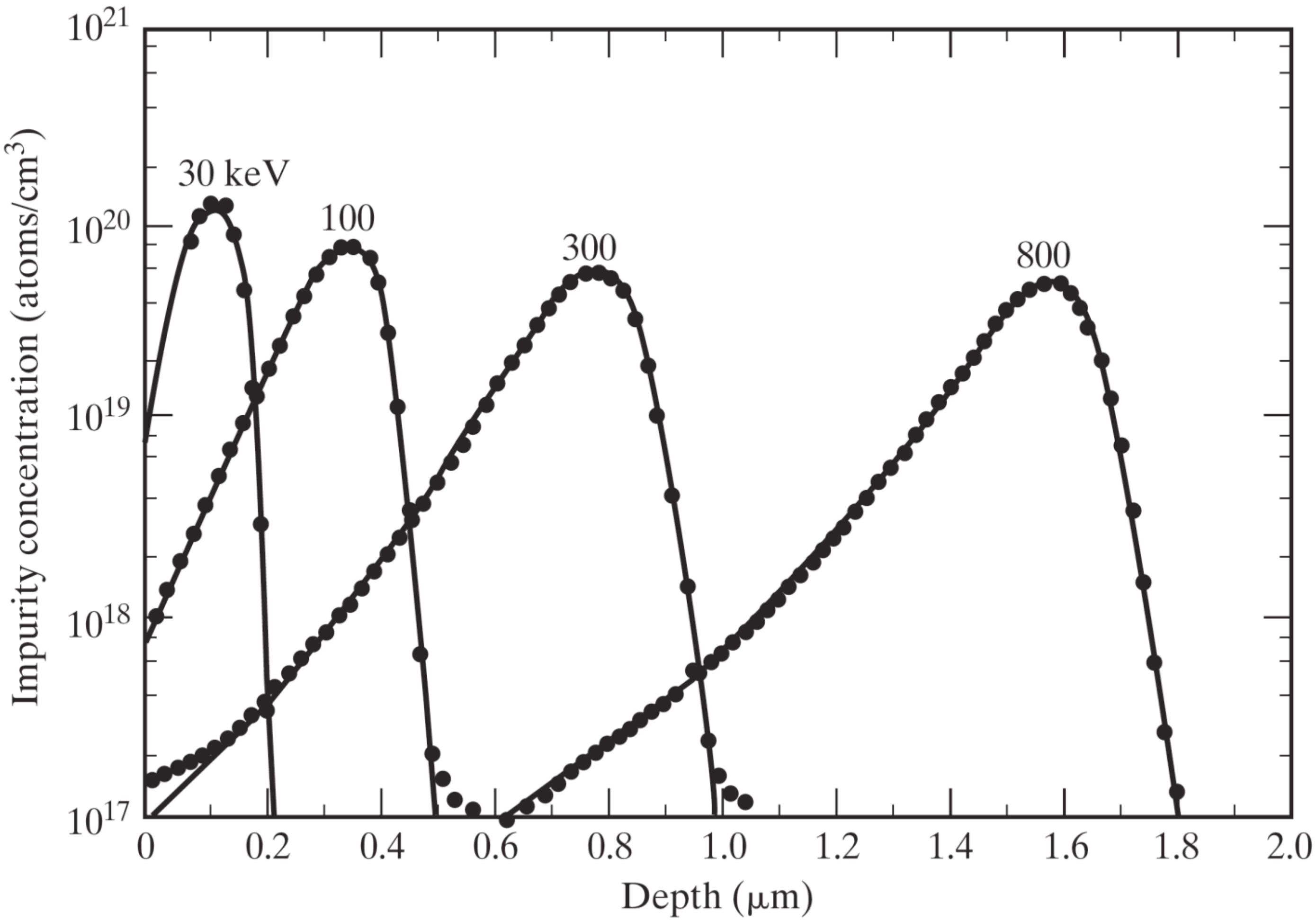
2. Place dopants into Si substitutional sites for electrical activation

Shallow Implantation



(c)

Deeper implantation



Curves deviate from Gaussian for deeper implants (> 200 keV)

5.5.5 Multiply Charged Ions

With Accelerating Voltage = x kV

Singly charged

H^+ B^+ As^+ → Kinetic Energy = x · keV

Doubly charged

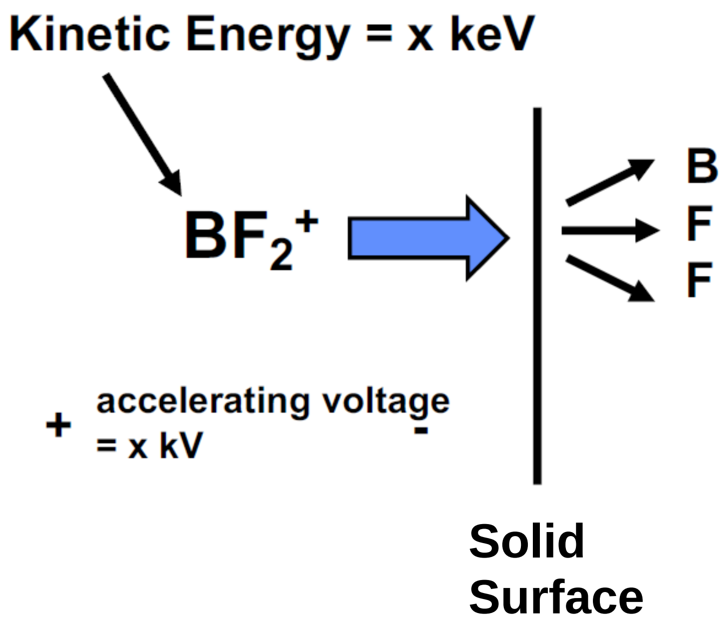
B^{++} → Kinetic Energy = 2x · keV

Triply charged

B^{+++} → Kinetic Energy = 3x · keV

Note:
Kinetic energy is expressed in eV. An electronic charge q experiencing a voltage drop of 1 Volt will gain a kinetic energy of 1 eV

5.5.5 Molecular ion implantation



Molecular ion will dissociate immediately into atomic components after entering a solid.

***All atomic components will have same velocity after dissociation*.**

$$\text{Velocity} \quad v_B = v_F = v_F$$

$$\text{K.E. of B} = \frac{1}{2} m_B \cdot v_B^{\,2}$$

$$\text{K.E. of F} = \frac{1}{2} m_F \cdot v_B^{\,2}$$

$$\frac{\text{K.E. of B}}{\text{K.E. of BF}_2^{\,+}} \approx \frac{11}{11+19+19} = 20\%$$

B has 11 amu

F has 19 amu

Outline

5.6.1 Summary

- **Why is ion implantation needed ?**

Key to introduce, advantage, steps

- **Basics of ion implanter**

Schematics, components, basically physics

- **Fundamentals of ion implantation**

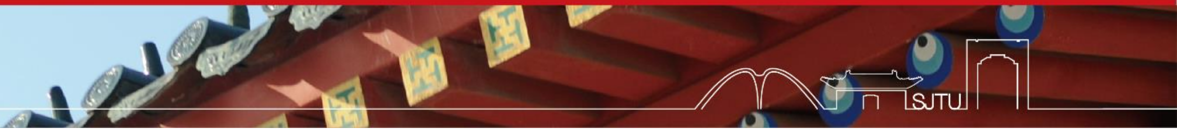
Ion stop mechanism, distribution model

- **Selective ion implantation**

Models, mask thickness, transmission factor of mask

- **Other aspect**

Junction depth, channeling effect and its reduction, Implantation damage, deeper and shallow implantation, multiple charged and molecular ion implantation



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